Spatial Estimation of Air Quality at Unmonitored Locations Using Machine Learning: Implications for IoT Sensor Network Design in Swedish Cities

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Popular Science Summary

Air pollution is one of the most well-documented threats to public health in cities, and Swedish cities are no exception. Two pollutants in particular, fine particulate matter (PM2.5) and nitrogen dioxide (NO2), are linked to breathing problems, heart disease, and premature death, even at the comparatively low concentrations typical of Sweden. To manage this risk, Sweden relies on a national network of official air quality monitoring stations, operated by SMHI, that measure these pollutants continuously and reliably. The problem is that these stations are expensive to build and maintain, so there are not very many of them, and they can be tens of kilometres apart. Between stations, nobody actually knows the air quality; it has to be estimated.

This thesis asks two related questions. First, how good can that estimate be? Using five years of data (2020 to 2024) from SMHI's stations, together with information about local land use, terrain, weather, and population density, a machine learning model was trained to estimate daily PM2.5 and NO2 concentrations at locations where no monitoring station exists. To test this fairly, each station was in turn held out and treated as if it had no data, and the model had to estimate its concentrations using only information from the other stations. This is a stricter test than it might sound, because nearby stations are excluded too, so the model cannot simply copy its closest neighbour's readings.

Second, this thesis asks how far away from a real station this kind of estimate can still be trusted. Intuitively, estimates should get worse the further they are from an actual measurement, and this thesis shows that this is indeed the case, especially for PM2.5, which behaves more smoothly across space than the sharply localised NO2 pollution found near traffic. By fitting a curve to how the estimation error grows with distance, it becomes possible to say, with reasonable confidence, up to what distance an estimate is still good enough to be useful, and beyond which a real sensor is genuinely needed.

That distance threshold is the key ingredient for the thesis's practical contribution: a data-driven method for recommending where to place new low-cost IoT air quality sensors across Sweden. Rather than choosing locations based on convenience or guesswork, this thesis uses the estimation-accuracy threshold to identify the parts of the country that are currently the most poorly served by existing monitoring, and then recommends, one sensor at a time, the locations that would do the most to close that gap. Applying this method nationally shows that a modest number of well-placed sensors can bring a substantially larger share of Swedish territory, including several currently unmonitored regions, within reliable reach of an air quality estimate, at a fraction of the cost of building more full-scale SMHI stations. The result is not a finished deployment plan, but a transparent, evidence-based starting point that city planners and environmental agencies can use to decide where additional monitoring would make the biggest difference.

Abstract

Sweden's national air quality reference network (SMHI) produces high-fidelity $PM_{2.5}$ and NO_2 measurements but is spatially sparse, leaving large areas without direct monitoring. This thesis investigates how accurately these pollutants can be estimated at unmonitored locations, how that accuracy degrades with distance from the nearest station, and how the resulting relationship can guide supplementary IoT sensor placement. Following a Design Science Research methodology, a Random Forest model trained on land-use, topographic, meteorological, and population covariates for 34 SMHI stations (2020–2024) was evaluated under buffered spatial leave-one-out cross-validation, benchmarked against inverse distance weighting (IDW) and land-use regression (LUR). For $PM_{2.5}$, the simplest baseline, inverse distance weighting, was the most accurate estimator (mean RMSE $2.33\,\mu g\,m^{-3}$, against $3.56\,\mu g\,m^{-3}$ for Random Forest), indicating that $PM_{2.5}$ concentrations are spatially smooth enough at the national scale that a distance-weighted average is difficult to improve upon with the available covariates. For NO_2, Random Forest achieved the lowest mean RMSE ($7.27\,\mu g\,m^{-3}$), outperforming both IDW and LUR. LUR was outperformed by both alternative methods for both pollutants. Decay curves fitted to per-station error against distance yielded reliable prediction thresholds of approximately 64 km ($PM_{2.5}$) and 6 km (NO_2). Applying the $PM_{2.5}$ threshold, a greedy sequential placement algorithm identified 20 candidate sensor locations, raising $PM_{2.5}$ land coverage from 10.9 % to 67.2 % and urban coverage from 38.8 % to 71.4 %, extending coverage to four of Sweden's seven unmonitored regions. The contribution is an empirically grounded, reproducible artefact translating spatial estimation accuracy into prioritised sensor placement guidance for Swedish municipal air quality planning.

Declaration of AI Usage

During the preparation of this thesis, an AI assistant (Claude, Anthropic) was used throughout the research and writing process. Its use included organising and tracking the project's research phases and task list; drafting and revising sections of the manuscript based on the author's research decisions, data, and results; writing and debugging data-processing and analysis code; and assisting with LaTeX formatting, compilation, and error resolution. All AI-generated content was reviewed, verified, and edited by the author. All research questions, methodological choices, data collection, analysis, and interpretation of results are the author's own work. Academic responsibility for the thesis in its entirety rests with the author. No AI tool was used to fabricate, alter, or substitute for empirical data or results.

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List of Abbreviations

Abbreviation	Full form
AIC	Akaike Information Criterion
API	Application Programming Interface
AQ	Air Quality
BIC	Bayesian Information Criterion
CORINE	Coordination of Information on the Environment (land cover dataset)
COVID-19	Coronavirus Disease 2019
CSV	Comma-Separated Values
DEM	Digital Elevation Model
DSR	Design Science Research
EDA	Exploratory Data Analysis
EPSG	European Petroleum Survey Group (coordinate reference system registry)
ESCAPE	European Study of Cohorts for Air Pollution Effects
ETRS89	European Terrestrial Reference System 1989
EU	European Union
GEOSTAT	Eurostat Geostatistical Population Grid
IDW	Inverse Distance Weighting
IoT	Internet of Things
JRC	Joint Research Centre (European Commission)
LUR	Land-Use Regression
MAE	Mean Absolute Error
ML	Machine Learning
NO_2	Nitrogen Dioxide
$PM_{2.5}$	Particulate Matter with diameter ≤ 2.5 μm
PSO	Particle Swarm Optimization
R^2	Coefficient of Determination
RF	Random Forest
RMSE	Root Mean Squared Error
RQ	Research Question
SLOO	Spatial Leave-One-Out (cross-validation)
SMHI	Swedish Meteorological and Hydrological Institute
WHO	World Health Organization

1 Introduction

Urban air quality is a measurable public health concern across European cities. Ground-level concentrations of fine particulate matter ($PM_{2.5}$) and nitrogen dioxide (NO_2) are among the most well-documented contributors to respiratory and cardiovascular morbidity, with exposure linked to premature mortality, reduced lung function, and elevated rates of hospital admission. Swedish cities are not exempt from this burden: studies of Stockholm, Gothenburg, and Umeå have demonstrated that traffic-related $PM_{2.5}$ and NO_2 exposure contributes significantly to attributable mortality and disease burden even under a comparatively strict national regulatory environment [1].

The fundamental challenge in monitoring these pollutants is spatial. Both $PM_{2.5}$ and NO_2 exhibit substantial intra-urban heterogeneity: concentrations vary significantly between urban cores, near-road zones, and suburban residential areas, reflecting the influence of localised emission sources such as traffic [2]. Characterising this spatial variability requires monitoring infrastructure at commensurate resolution. The Swedish Meteorological and Hydrological Institute (SMHI) operates the national network of regulatory-grade reference monitoring stations that constitutes the primary empirical record of air quality across Sweden. These stations produce high-fidelity measurements conforming to EU reference methods, but they are sparse by design: positioned to characterise regional background concentrations and to satisfy national reporting obligations, they are separated by margins that can span tens of kilometres. The areas between reference stations remain, by definition, unmonitored.

This spatial gap has practical consequences that extend beyond research. For environmental agencies and municipalities seeking to understand exposure at street or neighbourhood scale, the sparse reference network provides insufficient spatial resolution. For engineers and planners seeking to deploy supplementary monitoring infrastructure, the gap raises a more specific and largely unresolved question: where should sensors be placed to contribute information that is not already available from existing infrastructure? Recent years have seen rapid growth in Internet-of-Things (IoT) sensor networks based on low-cost electrochemical and optical instruments, which can be deployed at a fraction of the cost of regulatory-grade stations [3, 4]. Yet despite this technological availability, the placement of such networks is frequently driven by property access, convenience, or local intuition rather than any systematic analysis of where monitoring is genuinely needed. Without a principled basis for distinguishing locations where spatial estimation from existing infrastructure is adequate from locations where a physical sensor is warranted, IoT deployments risk duplicating coverage in accessible but already-monitored areas while leaving genuine blind spots unaddressed.

The problem this thesis addresses therefore sits at the intersection of spatial estimation and deployment decision-making. If the accuracy of model-based estimation from a reference station network can be quantified as a function of distance from the nearest station, that relationship provides an empirically grounded criterion for IoT deployment decisions. Translating this criterion into an actionable placement tool constitutes the research artefact developed here, positioned within a Design Science Research methodology [5, 6].

1.1 Research Aim and Research Questions

This thesis investigates how accurately air quality can be estimated at unmonitored locations using the SMHI reference station network and associated spatial covariates, how estimation accuracy varies with distance from the nearest reference station, and how the resulting empirical relationship can inform the systematic placement of IoT sensor nodes in a Swedish metropolitan case study. The research is structured around three questions:

1. **RQ1:** How accurately can machine learning models estimate daily $PM_{2.5}$ and NO_2 concentrations at spatially withheld SMHI reference stations using land-use, topography, and meteorological covariates?
2. **RQ2:** How does spatial estimation accuracy degrade as a function of distance from the nearest active reference station, and what is the maximum reliable prediction distance threshold before physical sensor deployment becomes necessary?
3. **RQ3:** Based on the accuracy-distance relationship modelled from the national network, how can optimisation heuristics be applied to guide the placement of IoT sensor nodes across Sweden, prioritising regions where estimation uncertainty is highest?

1.2 Scope and Delimitations

This thesis focuses on two pollutants: $PM_{2.5}$ and NO_2. These were selected for their distinct physical and chemical dispersion behaviours, their status as primary regulatory targets under EU Ambient Air Quality Directive thresholds and Swedish national standards, and their availability in the SMHI monitoring record. The temporal scope covers 2020 to 2024, a five-year window that encompasses variation in meteorological conditions and includes the COVID-19 period of reduced traffic activity in 2020 and 2021, which will be acknowledged and addressed in the data analysis.

The spatial scope for both model training and the placement analysis (RQ3) encompasses the full Swedish SMHI reference station network. The placement artefact is applied nationally: the data feasibility audit identified that 7 of Sweden's 21 administrative regions currently lack any reference station measuring both $PM_{2.5}$ and NO_2, making the national coverage area the natural domain for identifying where supplementary monitoring is most needed. No real-world sensor deployment is conducted as part of this study; the placement output is a data-driven research artefact and recommendation, not an operational infrastructure plan.

This study is restricted to daily average concentrations of $PM_{2.5}$ and NO_2; sub-daily dynamics, peak episode characterisation, and real-time forecasting are outside scope. The modelling framework is designed to characterise long-run spatial exposure patterns, not short-term pollution episodes. Indoor air quality, occupational exposure assessment, and atmospheric chemistry modelling are outside scope.

1.3 Thesis Structure

The remainder of this thesis is organised as follows. Section 2 reviews the relevant literature across urban air quality monitoring, spatial estimation methods, IoT sensor networks, and sensor placement optimization, and identifies the research gaps this work addresses. Section 3 presents the research methodology, including the Design Science Research framing, the data and validation design, and the justification for all methodological choices. Section 4 describes the data sources and feature engineering process. Sections 5 and 6 present the spatial estimation results and the accuracy-distance analysis, addressing RQ1 and RQ2 respectively. Section 7 describes the IoT placement artefact and the national case study, addressing RQ3. Section 8 discusses findings in relation to existing work, limitations, and implications for urban IoT planning. Section 9 concludes with a summary of contributions and directions for future research.

2 Background and Related Work

2.1 Urban Air Quality Monitoring and Regulatory Networks

Air quality monitoring in Europe operates through a hierarchy of infrastructure. At the highest level sit regulatory reference stations, operated by national meteorological and environmental agencies, which produce measurements conforming to EU reference methods and form the basis for national reporting under the Ambient Air Quality Directive (2008/50/EC). In Sweden, this function is fulfilled by SMHI, which operates the Luftwebb network of fixed monitoring stations measuring $PM_{2.5}$, NO_2, and other regulated pollutants at locations distributed across the country.

The health rationale for monitoring these pollutants is well established in the Swedish context. Segersson et al. [1] demonstrate that $PM_{2.5}$ and NO_2 from traffic and combustion sources contribute substantially to attributable cardiovascular and respiratory mortality in Stockholm, Gothenburg, and Umeå, at concentrations that may remain formally below EU limit values. Research on high-resolution pollutant dispersion in Swedish metropolitan areas has further shown that intra-urban exposure gradients are substantial, with marked differences between urban cores, near-road zones, and suburban residential areas [2].

These findings highlight a fundamental limitation of sparse reference networks. Regulatory stations are sited to represent broad background zones rather than local exposure conditions, and the spatial resolution they provide is insufficient for street-level or neighbourhood-level characterisation. Large areas between reference stations have no direct measurement record; concentrations at those locations must be estimated from the network, and the uncertainty in those estimates grows as the distance to the nearest station increases. Quantifying and managing this uncertainty is a central challenge for both public health assessment and supplementary infrastructure planning.

The limitations of sparse regulatory monitoring are increasingly discussed not in isolation, but in relation to the broader smart city paradigm, in which air quality is treated as one of several environmental and infrastructural parameters, alongside domains such as traffic, energy, and public safety, that a city is expected to sense, analyse, and act upon continuously. Munera et al. [7] conducted a systematic mapping study of 55 IoT-based air quality monitoring proposals developed explicitly within this smart city framing, and found that although technical activity in the field is substantial, integration between air quality sensing and the wider smart city data and decision-making ecosystem remains uneven, with monitoring capability often outpacing its use in actual planning processes. This suggests that the value of an air quality monitoring architecture, whether regulatory or IoT-based, depends not only on measurement accuracy but on whether its output can be translated into decisions that a city can act on, including where to place additional monitoring capacity. The present thesis addresses this translation problem directly, situating spatial estimation accuracy and IoT sensor placement within the same decision-oriented logic that motivates the smart city framing of urban air quality.

2.2 Low-Cost IoT Sensor Networks for Urban Monitoring

The emergence of low-cost air quality sensors based on electrochemical, optical, and metal-oxide technologies has opened the possibility of monitoring at spatial resolutions that are infeasible for regulatory-grade infrastructure. Karinthi et al. [3] provide a comprehensive review of low-cost sensor deployments, documenting capabilities across hardware platforms, deployment configurations, and calibration strategies. Laha et al. [8] further document how IoT architectures enable real-time environmental monitoring in outdoor urban settings, demonstrating that distributed microsensor networks can capture spatial variability that sparse reference networks cannot resolve. Popescu and Ionescu [4] demonstrate this capability in practice: IoT deployments in urban environments yield spatially rich records of PM and gaseous pollutant patterns that complement the regulatory station baseline.

A consistent finding across this literature is that low-cost sensor networks face a placement problem that technology alone does not resolve. The technical feasibility of dense monitoring does not determine where monitoring is needed. In the absence of an analytical basis for identifying which locations benefit most from physical measurement, deployments remain subject to practical constraints such as property access, power availability, and the preferences of hosting institutions, rather than the spatial structure of pollution exposure. The literature confirms that IoT networks can provide valuable supplementary data, but offers limited guidance on the criteria by which placement decisions should be made.

2.3 Spatial Estimation of Air Quality at Unmonitored Locations

Estimating air quality at locations without physical measurements is a long-standing problem in environmental research. Approaches span a range of complexity and theoretical assumptions, from deterministic interpolation to regression modelling to machine learning, and each carries distinct implications for accuracy, interpretability, and data requirements.

The simplest interpolation approach, inverse distance weighting (IDW), estimates concentrations at unmonitored locations as a distance-weighted average of observations at nearby reference stations. IDW makes no assumptions about the mechanisms driving spatial variation and performs poorly in environments where pollution is structured by land-use patterns, topography, or meteorological conditions rather than simple proximity. It nonetheless remains a standard deterministic baseline against which more sophisticated methods are evaluated.

Land-use regression (LUR) represents the established peer-reviewed standard for predicting long-run outdoor pollutant concentrations at unmonitored locations in European contexts. LUR models relate observed concentrations at reference stations to predictor variables derived from geographic data: road density, traffic volume, land cover classification, population density, and similar covariates measured within buffer zones around each station. The resulting regression equation is then applied to unmonitored locations where the predictor variables are available. Hoek et al. [9] review 25 LUR studies and establish the approach as the dominant framework for intra-urban exposure modelling in Europe. The European Study of Cohorts for Air Pollution Effects (ESCAPE), a multi-centre European project (2008–2012) that pooled exposure and health data from more than thirty existing cohort studies to assess the long-term health effects of air pollution, subsequently validated LUR across 20 European study areas using a standardised modelling and evaluation protocol: Eeftens et al. [10] reported cross-validation R^2 values of approximately 0.71 for $PM_{2.5}$ models, while Wang et al. [11] reported leave-one-out cross-validation R^2 of 0.83 for NO_2 models across the same areas. These benchmarks represent the standard against which any alternative spatial estimation approach operating in a comparable European data environment must be evaluated.

More recent Swedish research has characterised intra-urban exposure gradients using high-resolution dispersion modelling across Stockholm, Gothenburg, and Malmö, confirming that the spatial heterogeneity documented in ESCAPE-era studies persists in contemporary Swedish cities [2]. The ESCAPE R^2 benchmarks therefore remain the relevant standard for evaluating any spatial estimation framework operating on the national SMHI station network.

LUR has known limitations. Model performance can degrade outside the range of predictor values observed during training, which is a constraint when applying a nationally-trained model to locations with unusual land-use configurations. The linear regression framework may also fail to capture non-linear interactions between predictors present in complex urban environments.

Machine learning (ML) methods have been increasingly applied to spatial air quality estimation, motivated by their capacity to model non-linear relationships between predictor variables and pollutant concentrations. Agbehadji and Obagbuwa [12] provide a systematic review of ML and deep learning applications to spatiotemporal air quality prediction, documenting a consistent pattern of ML models outperforming traditional regression baselines on RMSE and MAE metrics across diverse geographic contexts. Chen and Lin [13] demonstrate ML-based spatial interpolation for $PM_{2.5}$ estimation, integrating microsensor cluster data with a regulatory station baseline to achieve high spatial resolution estimates across an urban area. Hybrid approaches combining ML feature extraction with geostatistical variance modelling have also been proposed: Xie et al. [14] integrate a deep learning temporal encoder with Kriging spatial covariance estimation, demonstrating accuracy improvements in data-sparse regions. The applicability of such hybrid approaches in the present study is conditional on the national station count confirmed in the data feasibility phase.

A methodological concern that runs throughout the spatial ML literature is spatial autocorrelation. When training data are spatially autocorrelated, standard random cross-validation systematically over-estimates out-of-sample accuracy: nearby stations share similar predictor environments and similar concentration values, so a model trained on a station's neighbours has, in effect, partial knowledge of the target station's behaviour. Roberts et al. [15] demonstrate formally that standard cross-validation strategies under-estimate prediction error in spatially structured data, and recommend block-based or location-withholding strategies as more honest evaluations of predictive performance. This concern is directly relevant to the validation design for any estimation framework trained on a reference station network, and is addressed in the methodology presented in Section 3.

2.4 Sensor Placement Optimisation

Given a model of spatial estimation uncertainty across an area, the problem of where to place physical sensors can be framed as an optimisation problem: locate a fixed number of sensors to minimise residual uncertainty or to maximise spatial coverage across the target domain. Several algorithmic families have been applied to this problem.

Particle Swarm Optimization (PSO) is a population-based metaheuristic that maintains a set of candidate solutions, or particles, which move through the solution space guided by each particle's individual best position and the globally best position found across the swarm. Gupta and Thakur [16] apply PSO to outdoor air quality sensor placement, treating the monitoring area as a grid and optimising node positions to reduce network-wide estimation variance. Their study demonstrates the applicability of the approach to combinatorial outdoor placement problems. Enhanced PSO formulations that improve convergence and spatial coverage for wireless sensor network deployment have been proposed in related work [17].

The placement studies reviewed above operate at single-area or small-scale experimental scope rather than across a full national reference network: Gupta and Thakur [16] optimise placement within a single monitoring area treated as a grid, and the enhanced PSO formulation of [17] addresses wireless sensor network deployment without reference to distance-based estimation accuracy from an existing monitoring infrastructure. Outdoor urban sensor network design at the scale of a metropolitan region, grounded in an empirically derived model of estimation error as a function of distance from existing reference infrastructure, remains relatively unexplored. This gap motivates the placement artefact developed in this thesis.

2.5 Research Gaps and Positioning

The literature reviewed in this section reveals three interconnected gaps that this thesis addresses.

First, while ML methods for spatial air quality estimation have been studied extensively, most evaluations use standard random cross-validation that does not adequately account for spatial autocorrelation. The accuracy achievable at genuinely unmonitored locations, assessed using a spatially honest validation protocol, is therefore less certain than headline performance figures suggest. RQ1 addresses this gap by evaluating spatial estimation accuracy under a geographically rigorous validation design.

Second, the relationship between spatial estimation accuracy and the distance from the nearest reference station has not been systematically quantified across a national-scale monitoring network. Aggregate cross-validation performance figures do not characterise how performance varies with the spatial configuration of the network relative to a prediction target. Understanding this distance-accuracy relationship is a prerequisite for any principled deployment criterion. RQ2 addresses this gap by characterising the decay in estimation accuracy as a function of distance.

Third, while PSO-based sensor placement has been demonstrated in controlled settings, its application to outdoor IoT network design grounded in an empirically derived estimation error model has not been demonstrated for a Swedish urban context. The body of work on systematic outdoor placement in environments with well-characterised reference networks is limited. RQ3 addresses this gap by developing a placement artefact anchored in the empirical results of RQ1 and RQ2 and applied to a Swedish metropolitan case study.

Section 3 presents the methodological approach taken to address each of these gaps.

3 Research Methodology

3.1 Research Design and Paradigm

This thesis adopts a Design Science Research (DSR) paradigm, following the methodology proposed by Peffers et al. [5] and grounded in the principles for rigorous design science articulated by Hevner et al. [6]. DSR is appropriate for research whose primary contribution is a designed artefact: a method, model, or tool that addresses a real-world problem and is evaluated against real-world criteria. The choice of DSR is not merely a framing convenience. The three research questions of this thesis collectively produce outputs of two distinct types: empirical findings (the estimation accuracy characterisation for RQ1 and RQ2) and a designed object (the sensor placement procedure for RQ3). DSR provides the methodological vocabulary that accommodates both within a single coherent inquiry, and requires that the artefact be evaluated not only by internal model metrics but by how well it addresses the real-world problem that motivated the research.

The Peffers et al. [5] framework prescribes six phases of design science inquiry. Table 2 maps each phase to this thesis.

Table 2: Mapping of the Peffers et al. (2007) DSR phases to this thesis.

DSR phase	**This thesis**
1. Problem identification and motivation	The SMHI Luftwebb network leaves large intra-urban areas without direct air quality measurement. IoT deployments lack analytical criteria for placement decisions. The health burden of $PM_{2.5}$ and NO_2 exposure in Swedish cities motivates better spatial characterisation of pollutant concentrations.
2. Define objectives of a solution	RQ1 specifies the target: an ML estimation model whose out-of-sample accuracy at withheld stations is characterised under a spatially honest validation protocol. RQ2 specifies the distance-accuracy decay relationship and the reliable prediction distance threshold. RQ3 specifies the placement artefact: a prioritised set of IoT sensor deployment coordinates for a Swedish metropolitan case study, grounded in the empirical results of RQ1 and RQ2.
3. Design and development	A spatial estimation framework is constructed using a Random Forest model with land-use, topographic, and meteorological covariates, evaluated against IDW and land-use regression (LUR) benchmarks under a buffered spatial leave-one-out cross-validation protocol. The per-station error-distance relationship is modelled to produce a decay curve and a threshold. A greedy sequential placement algorithm operationalises the threshold as a spatial coverage objective over the national territory.
4. Demonstration	The placement artefact is applied to Sweden nationally as the case study, producing a prioritised list of IoT sensor deployment coordinates across all regions alongside before-and-after coverage uncertainty maps.
5. Evaluation	The estimation model is evaluated by RMSE, MAE, and R^2 per pollutant, benchmarked against IDW and LUR. The placement artefact is evaluated by the proportion of Swedish urban area brought within reliable prediction distance of at least one sensor (existing or newly placed), and by the number of currently unmonitored regions transitioning to served, as the primary real-world criteria.
6. Communication	Findings are presented in this thesis, with methodology, empirical results, and artefact design reported in sufficient detail for replication and extension.

3.2 Spatial Estimation Strategy

3.2.1 Pollutant Separation

$PM_{2.5}$ and NO_2 exhibit physically distinct spatial behaviours that preclude joint modelling. NO_2 is primarily a traffic-sourced pollutant whose concentration gradients are sharp in the vicinity of road infrastructure, decaying rapidly with distance from source; the spatial range of strong autocorrelation is typically on the order of 500 metres to 2 kilometres in European urban areas. $PM_{2.5}$ has a larger spatial footprint resulting from mixed source contributions including regional transport, secondary formation, and diffuse combustion; concentration gradients are smoother and the spatial range of autocorrelation typically extends to 5–20 kilometres. Accordingly, separate models are constructed, validated, and analysed for each pollutant throughout this study. Separate decay thresholds are expected from the accuracy-distance analysis (Section 3.4), and all results tables in Sections 5 and 6 present $PM_{2.5}$ and NO_2 independently.

3.2.2 Feature Set

The feature set comprises 30 predictor variables spanning five groups. Land-use covariates are derived from the CORINE Land Cover 2018 dataset, expressed as proportions of nine functionally grouped land-use classes within buffer radii of 500 m and 1 km around each station, following the buffer geometry established as standard in European land-use regression studies [9]. Road and rail network proximity is captured within this land-use group via the proportion of CORINE class 122 within each buffer, reflecting the 100 m resolution of the raster source and removing the need for a separate road-length covariate. This yields 18 land-use proportion features. Topographic covariates include station elevation and terrain roughness within a 1 km radius from the EU-DEM (two features). Population density is captured by gridded population counts within 1 km and 5 km radii (two features). Meteorological covariates include daily wind speed, wind direction, air temperature, relative humidity, and precipitation (five features). Station latitude and longitude are included as spatial position features (two features). A binary indicator for the 2020–2021 COVID-19 lockdown period accounts for the structural break in traffic-related emissions confirmed during exploratory analysis of the study-period concentration time series (one feature).

The resulting feature count of 30 exceeds the 10–15 target stated at the design stage, reflecting the decision to retain all 18 CORINE proportions at both buffer scales rather than selecting a subset. With 11 unique spatial training locations, the ratio of features to independent observations is high by the standards of linear modelling. Random Forest is well-suited to this regime: its bootstrap aggregation and random subsampling of predictors at each split act as implicit regularisation, averaging out spurious associations without requiring a separate variable selection procedure. No explicit feature selection step was applied. The data feasibility audit confirmed 11 stations passing the completeness filter concurrently for both pollutants, consistent with individual ESCAPE study areas that typically comprised 20–40 monitoring sites per urban region [10, 11].

3.2.3 Model Selection

Three estimation approaches are implemented for each pollutant. Inverse distance weighting (IDW) serves as the deterministic spatial baseline, requiring no covariates and relying solely on proximity to observed stations. It provides a lower performance bound against which the contribution of predictor information can be assessed.

Land-use regression (LUR), calibrated on the SMHI station data using the predictors described above, serves as the established European peer-reviewed benchmark. The ESCAPE cross-validation R^2 values reported in Section 2, approximately 0.71 for $PM_{2.5}$ [10] and 0.83 for NO_2 [11], represent the standard that any ML approach operating in a comparable European data environment must meet or exceed to demonstrate added value over the existing methodological baseline.

Random Forest (RF) is selected as the primary machine learning model. This selection is supported by four considerations. First, RF is well-established in the spatial air quality estimation literature, demonstrating consistent accuracy improvements over LUR across diverse geographic and data environments [12, 13]. Second, RF handles mixed feature types natively, combining categorical land-use proportions, continuous meteorological variables, and topographic covariates without requiring distributional assumptions about the response. Third, with a sparse training network of 11 unique spatial locations, a model that makes no distributional assumption about the joint feature-concentration relationship and is robust to irrelevant predictors is preferable to approaches that impose stronger structure. Fourth, the bootstrap ensemble mechanism of RF reduces variance without requiring explicit regularisation, which is advantageous when training locations are too few to support extensive hyperparameter tuning. Deep learning architectures and hybrid geostatistical approaches such as those reviewed in Section 2 are excluded from this study: such models require substantially more unique spatial training locations than the SMHI network is expected to provide, and their additional complexity is not warranted for the contribution claimed here.

3.3 Spatial Validation Protocol

Roberts et al. [15] demonstrate that standard random cross-validation produces optimistically biased error estimates for spatially autocorrelated data. When training and test observations are spatially proximate, nearby training stations share similar predictor environments and concentration levels with the withheld station; the model therefore has partial information about the test target via its training data, and the resulting performance estimate does not reflect true out-of-sample generalisation to unmonitored locations.

This study addresses the spatial autocorrelation problem through a buffered spatial leave-one-out cross-validation (SLOO) protocol [18]. In each fold, a single reference station is withheld as the test location. In addition, all stations within a minimum distance d_{buffer} of the withheld station are excluded from the training set, preventing the model from leveraging the spatial proximity of near neighbours to the test point. The remaining stations are used for training, and a prediction is made at the withheld location. The procedure is repeated for every station, producing a complete set of spatially honest out-of-sample prediction errors.

The exclusion buffer radius d_{buffer} is set to 5 km, following the inter-station distance audit reported in Appendix A. The 34-station network contains 24 station pairs separated by less than 5 km, all corresponding to within-city clusters in Stockholm, Uppsala, Norrköping, Västerås, and Malmö. A buffer that fails to exclude these pairs allows the model to exploit spatial proximity within urban clusters rather than demonstrating genuine generalisation to unmonitored locations. A 5 km buffer removes all 24 such pairs and leaves a minimum of 26 training stations in every fold.

The empirical variogram range of 150–200 km, which would represent the theoretically optimal buffer following Roberts et al. [15], is operationally infeasible: the most isolated station in the network has a nearest neighbour at 81.9 km, so a buffer of that magnitude would leave it with no training data. The 5 km buffer is therefore justified by network structure rather than spatial autocorrelation range; this constraint is acknowledged in Section 6. The buffered SLOO protocol is applied separately for $PM_{2.5}$ and NO_2.

Standard random k-fold cross-validation is explicitly excluded. Random partitioning does not guarantee spatial independence between training and test sets in a geographically distributed station network, and any performance figure derived from such a protocol cannot be interpreted as an estimate of accuracy at genuinely unmonitored locations [15].

3.4 Accuracy-Distance Analysis

The per-station prediction errors produced by the buffered SLOO procedure serve as the primary input to the accuracy-distance analysis for RQ2. For each withheld station, the root mean squared error (RMSE) is recorded alongside the Euclidean distance from the withheld station to its nearest retained training station at the time of prediction, that is, the nearest station lying outside the exclusion buffer. Euclidean distance is used throughout; for NO_2, road-network distance would be more physically meaningful given the traffic-source dominance of that pollutant, but the sparsity of the SMHI network precludes robust road-network distance computation, and Euclidean distance provides a consistent and reproducible metric across both pollutants. This distance quantifies the minimum spatial gap the model must bridge from observed data.

The resulting set of (distance, error) pairs is used to fit three candidate functional forms: logarithmic, power-law, and exponential decay. Form selection is performed by the Akaike Information Criterion (AIC). The selected function provides a continuous mapping from distance to expected prediction error. A reliable prediction distance threshold is defined as the distance at which expected RMSE reaches a criterion bound; the specific bound is determined from the EDA phase (Section 4), informed by the empirical pollutant concentration distributions and the tolerances implied by the WHO Air Quality Guidelines for $PM_{2.5}$ and NO_2 respectively. The analysis is conducted separately for each pollutant, and separate thresholds are reported.

A potential confound must be acknowledged: stations with small inter-station distances are predominantly classified as urban traffic stations, co-located near road infrastructure within the same city, while stations with large inter-station distances include both urban background and rural background monitors. The observed error-distance relationship may consequently reflect both genuine distance-based accuracy degradation and systematic differences in the complexity of local pollutant gradients between traffic-dominated and background environments. To assess this confound, a sensitivity analysis stratifies the 11 analytical stations by SMHI monitoring classification (urban traffic versus urban background and rural background), and separate decay curves are fitted for each stratum. Where the stratum-specific curves are consistent with the pooled result, the pooled analysis is considered robust; where they diverge materially, stratum-specific thresholds are reported and the implications for deployment guidance are discussed.

The study period 2020–2024 encompasses the COVID-19 pandemic, during which traffic volumes and industrial activity in Sweden were reduced, particularly in 2020 and 2021. This temporal discontinuity may have altered the feature-to-concentration relationships that the model is trained to capture. Two strategies address this. First, a binary covariate `covid_period` (1 = 2020–2021, 0 = 2022–2024) is included in the feature matrix, allowing the Random Forest model to learn any systematic offset associated with the pandemic period rather than treating those years as structurally normal. Second, a sensitivity analysis re-runs the full estimation and decay analysis restricted to the 2022–2024 sub-period. If the decay curves derived from the two periods are consistent, the full 2020–2024 model is reported as the primary result; if they diverge materially, the 2022–2024 model is reported as primary and the COVID-period effect is discussed in Section 8.

3.5 Sensor Placement Optimisation

The placement optimisation addresses RQ3 by identifying candidate IoT sensor locations that would most efficiently extend the reliable coverage of the existing reference network across Sweden. An uncertainty surface is constructed over the national territory by computing, for each cell of a regular spatial grid, the distance to the nearest existing reference station. Grid cells whose nearest-station distance exceeds the reliable prediction distance threshold derived in Section 3.4 are classified as underserved by the current network.

The placement procedure is a greedy sequential algorithm. At each step, the candidate location whose addition to the sensor set would maximally reduce the total underserved area is selected. The procedure produces a ranked list of deployment coordinates, with each successive location providing the greatest marginal coverage gain over what preceding placements have already achieved. Placement continues until a specified number of sensors has been added or until the residual underserved area falls below a defined tolerance.

Greedy sequential placement is preferred over population-based metaheuristic approaches such as Particle Swarm Optimization for three reasons. First, the spatial coverage objective defined here is a well-studied combinatorial problem for which greedy search is known to perform within a constant factor of the global optimum for coverage-type objectives; the approximation quality is adequate for the practical planning purpose of this artefact. Second, the DSR evaluation phase [5] requires that the artefact be evaluable by practitioners: a placement procedure whose decisions are transparent and reproducible step by step is more useful as a planning tool than one derived from a population-based search whose convergence and parameter sensitivity require specialist interpretation. Third, confirming that the coverage landscape is sufficiently multimodal to warrant PSO, a prerequisite for any metaheuristic advantage over greedy search [16, 17], is not possible prior to the construction of the uncertainty surface in Section 7.1. The greedy approach is locally rather than globally optimal: early placements constrain subsequent ones. This limitation is acknowledged, and the resulting placement recommendations are presented as an evidence-based starting point for municipal planning, not as a provably optimal sensor configuration.

The output of this artefact is a prioritised set of candidate coordinates for further evaluation by municipal planners; it does not constitute a deployment specification. Infrastructure constraints such as power availability, property access, mounting requirements, and hardware selection are outside the scope of this study and must be resolved through site-level assessment. A population-based approach such as Particle Swarm Optimization [16] may offer improvements in global optimality once the uncertainty surface has been established from the results in Section 7.1, and represents a natural direction for future research.

3.6 Evaluation Metrics

Table 3 summarises the evaluation metrics applied at each stage of the study.

Table 3: Evaluation metrics by research question and phase.

Metric	Applies to	Rationale
RMSE ($\mu g\, m^{-3}$)	RQ1 (primary)	Penalises large errors more heavily; interpretable in concentration units; standard in spatial estimation literature
MAE ($\mu g\, m^{-3}$)	RQ1 (secondary)	Robust to outliers; complements RMSE; reported for completeness
R^2	RQ1	Required for comparison with ESCAPE LUR benchmarks reported in R^2 form [10, 11]
Reliable prediction distance threshold (km)	RQ2	Distance at which expected RMSE reaches the criterion bound; derived separately per pollutant
Proportion of urban area within threshold (%)	RQ3	Primary real-world DSR evaluation criterion; percentage of Swedish urban area within reliable prediction distance of at least one sensor, before and after placement
Regions transitioning to served (#)	RQ3 (secondary)	Number of Swedish regions currently without any station within threshold that would be served by the proposed placements

All metrics are computed separately for $PM_{2.5}$ and NO_2. Model performance is assessed against the IDW baseline, the LUR benchmark, and the ESCAPE R^2 standard values. A model that does not exceed the LUR benchmark under the buffered SLOO protocol does not demonstrate added value over the established European methodology.

3.7 Validity, Reliability, and Ethical Considerations

Validity. The internal validity of the estimation results rests on the spatial honesty of the validation protocol. The buffered SLOO design [18, 15] is specifically intended to prevent spatial leakage between training and test data, ensuring that reported error estimates reflect performance at genuinely unmonitored locations rather than at sites whose predictor environments are represented in the training set. The stratified sensitivity analysis for the decay curve provides a check on the confound between distance and network density. External validity is limited to the Swedish national station network and its geographic and meteorological context; generalisation to other national networks would require replication in those data environments.

Reliability. All data sources are publicly accessible and documented: SMHI Luftwebb, the SMHI meteorological observation API, CORINE Land Cover, and the national digital elevation model. The estimation, validation, decay analysis, and placement procedures are implemented in code with fixed random seeds for reproducibility. The full feature matrix, model configurations, and validation fold assignments are documented and retained for inspection.

Ethical considerations. All data sources used in this study are open-data products carrying no personal information. No individual-level records are processed at any stage. The study does not involve human participants and does not require ethical review under

Swedish research ethics legislation. The use of AI-assisted tools in the preparation of this thesis is disclosed in the Declaration of AI Usage.

4 Data and Feature Engineering

This study draws on five open-access data sources to construct the unified feature matrix used in the spatial estimation and placement analyses: SMHI air quality observations, SMHI meteorological covariates, CORINE Land Cover 2018, the EU Digital Elevation Model, and the Eurostat GEOSTAT 2018 population grid. All sources use the ETRS89 Lambert Azimuthal Equal-Area projection (EPSG:3035) as the common coordinate reference system, which preserves area measurements consistently across the Swedish study extent. The remainder of this section describes each source, the feature extraction procedures applied, and the data quality decisions made prior to modelling.

4.1 Air Quality Observations

Air quality measurements were obtained from SMHI Datavärdskap Luft, Sweden's national air quality data repository [19]. The data consist of daily mean surface concentrations of NO_2 and $PM_{2.5}$ at reference monitoring stations across Sweden. The study period spans 1 January 2020 to 31 December 2024, corresponding to 1,827 calendar days. Observations were downloaded in semicolon-delimited CSV format from the Luftwebb portal for all stations reporting at least one of the two target pollutants during this period.

A total of 34 stations were identified with $PM_{2.5}$ or NO_2 records in the study period. To ensure that spatial estimation models are trained on reliable time series, a completeness threshold of 90 per cent of daily records was applied per station and per pollutant. Nineteen of the 34 stations meet this threshold for at least one of the two target pollutants; of these, 11 meet it concurrently for both $PM_{2.5}$ and NO_2. Because $PM_{2.5}$ and NO_2 are modelled separately but evaluated under the same spatial validation design (Section 3.3) and compared directly in the accuracy-distance analysis (Section 6), the analytical sample is restricted to these 11 concurrently-passing stations: using a different station set per pollutant would confound genuine differences in accuracy-distance behaviour with differences in the geographic sampling of the two validation networks. These 11 stations span six Swedish regions: Stockholm (five stations), Uppsala (two stations), Västernorrland, Skåne, Kalmar, and Jämtland (one station each). The majority are classified as urban traffic or urban background stations. Two stations are non-urban: Bredkälen, classified as rural background, and Norr Malma in Norrtälje, classified as rural-regional background; these are the only two rural stations in the analytical sample, a constraint of the Swedish national monitoring network that is addressed in the accuracy-distance analysis (Section 6).

All 34 stations are retained in the feature matrix and may contribute daily training rows to the buffered spatial leave-one-out procedure described in Section 3.3. The 90 per cent completeness threshold governs which stations are eligible to be withheld and tested as validation targets, not which stations may appear in the training data for a given fold: only the 11 concurrently-passing stations qualify for this role. $PM_{2.5}$ and NO_2 are modelled and analysed separately throughout, reflecting their different spatial gradient characteristics and emission source profiles.

These 11 concurrently-passing stations form the spatial leave-one-out validation sample used throughout Sections 5 and 6. The remaining eight stations that pass the 90 per cent threshold for one pollutant only (seven for NO_2, one for $PM_{2.5}$) are not used as validation targets for either pollutant, for the reason given above.

4.2 Meteorological Covariates

Meteorological covariates were retrieved from the SMHI Open Data meteorological observation API [20]. For each of the 34 AQ stations, the nearest available SMHI surface observation station was identified for each of five parameters: air temperature (degrees Celsius), wind speed ($m \cdot s^{-1}$), wind direction (degrees), relative humidity (per cent), and precipitation (mm/day). Parameters were downloaded at hourly resolution and aggregated to daily values. Wind direction was aggregated using the circular mean of the sine and cosine components to handle the 0°/360° discontinuity. Precipitation was stored as daily totals in the source files and was carried forward without aggregation.

These five parameters capture the principal atmospheric drivers of near-surface pollutant transport and accumulation. Wind speed and direction govern advective dispersal and local plume dilution. Air temperature influences boundary layer mixing height and photochemical NO_2 formation. Relative humidity affects $PM_{2.5}$ hygroscopic growth and secondary aerosol partitioning. Precipitation is associated with wet deposition and pollutant scavenging.

Coverage varies across parameters. Wind speed, wind direction, and relative humidity were available at all 34 stations with greater than 95 per cent daily completeness. Air temperature was available at 31 stations with adequate coverage; the three remaining stations, two in Västerås and one in Varberg, are affected by the operational closure or inactivity of their nearest meteorological observation station during parts of the study period. Precipitation coverage was substantially lower: only 19 of 34 stations have complete daily records, with the remaining 15 stations affected by SMHI's phased transition from traditional precipitation gauges to a newer automated gauge network, which resulted in systematic gaps for 2020 to 2022 at these stations. The handling of missing precipitation values is described in Section 4.7.

4.3 Land-Use Features

Land-use composition around each station was characterised using the CORINE Land Cover 2018 raster product (version 2020_20u1) from the EU Copernicus Land Monitoring Service [21]. The raster is provided at 100 m spatial resolution in EPSG:3035 and covers the full extent of Sweden. Land-use buffer features at circular radii of 500 m and 1 km around each station are a standard input in European LUR modelling [9, 10, 11]; the same buffer scales are adopted here to maintain comparability with the ESCAPE benchmarks.

The 44 CORINE classes were aggregated into nine functionally meaningful groups: continuous urban fabric (class 111), discontinuous urban fabric (class 112), industrial and commercial areas (class 121), road and rail networks (class 122), ports and airports (classes 123 and 124), green urban areas (classes 141–142), agricultural land (classes 211–244), forest (classes 311–324), and water bodies and wetlands (classes 411–523). The proportion of each group within a circular buffer was computed for both the 500 m and 1 km radii, yielding 18 land-use proportion features per station. All proportions within each buffer radius sum to 1.0.

4.4 Topographic Features

Elevation data were derived from the EU-DEM version 1.1 mosaic produced by the European Environment Agency under the Copernicus programme [22]. The original mosaic covers Europe at 25 m resolution in EPSG:3035. To reduce storage requirements while retaining sufficient spatial detail for station-level extraction, the raster was resampled to 100 m resolution using spatial averaging and clipped to the Swedish national extent.

The resulting file covers an elevation range of −58 m to 2,415 m, with the upper range attributable to Norwegian mountain areas near the western border.

Two topographic features were extracted per station: (1) elevation at the station location in metres, obtained by bilinear interpolation from the resampled DEM, and (2) terrain roughness within a 1 km buffer, defined as the standard deviation of elevation values within the buffer. Terrain roughness serves as a proxy for local topographic complexity that may influence near-surface wind flow and pollutant dispersal patterns.

4.5 Population Density

Gridded population data were taken from the Eurostat GEOSTAT 2018 population grid, produced by the Joint Research Centre [23]. The grid provides population counts at 1 km^2 resolution in EPSG:3035. Two features were extracted per station: total population within a 1 km radius and total population within a 5 km radius. These two scales capture the immediate residential density around the station and the broader urban agglomeration, respectively. Population density is an established predictor in LUR models, reflecting the aggregate contribution of residential heating, traffic, and economic activity to local pollution concentrations [9].

4.6 Feature Matrix Construction

The five data sources were joined on station code and calendar date to produce a unified daily feature matrix covering all 34 stations and the full 2020–2024 study period. Date-station rows for which both NO_2 and $PM_{2.5}$ were simultaneously absent were excluded; all other combinations were retained to preserve as many training observations as possible. The resulting matrix contains 48,283 rows and 37 columns, as listed in Table 4.

A binary indicator variable `covid_period` was added: observations from 2020 and 2021 were assigned the value 1, and observations from 2022 to 2024 were assigned the value 0. This flag enables sensitivity analyses that exclude or stratify the COVID-19 period, during which reduced traffic volumes and industrial activity in Sweden are expected to have altered the feature-target relationships that models are trained to capture.

Table 4: Feature matrix column inventory (48,283 rows, 37 columns). CORINE proportions sum to 1.0 within each buffer radius.

Feature group	**Description**	**n**
Identification	date, station code, name, type, latitude, longitude	6
Period indicator	covid_period (1 = 2020–2021, 0 = 2022–2024)	1
Meteorological	temp_C, wind_speed_ms, wind_dir_deg, rh_pct, precip_mm, precip_observed	6
Land use (500 m)	9 CORINE groups at 500 m buffer	9
Land use (1 km)	9 CORINE groups at 1 km buffer	9
Topography	elevation_m, terrain_rough_m	2
Population density	pop_1km, pop_5km	2
Target variables	NO2_ugm3, PM25_ugm3	2
Total		**37**

4.7 Data Quality and Missing-Data Handling

Three data quality procedures were applied to the raw feature matrix before it was used in model development.

Negative concentration clipping. A small number of daily records contain negative concentration values arising from instrument drift and calibration adjustments in the SMHI reporting pipeline. Twelve such values were identified across the NO_2 and $PM_{2.5}$ columns. These values are physically impossible and were clipped to zero. Clipping rather than deletion was chosen to preserve the otherwise complete date-station record.

Precipitation imputation. As noted in Section 4.2, 15 of the 34 stations have systematic precipitation gaps for 2020–2022 attributable to this phased network transition. Across the full matrix, 15,236 daily precipitation values (31.6 per cent) were absent. These gaps were imputed using a monthly climatological mean: for each station, the mean observed precipitation was computed separately for each calendar month across all available years, and missing values were replaced with the corresponding station-month mean. Where no observed values existed for a given station-month combination, the national monthly median across all stations was used as a fallback. The binary column `precip_observed` indicates original measurements (1) versus imputed values (0), enabling downstream sensitivity assessment.

Station-level completeness filtering. The 11 concurrently-passing stations used in model training and validation were confirmed by checking completeness against the full 1,827-day study period. A further eight stations pass the 90 per cent threshold for one pollutant only and are excluded from the analytical sample for the reason given in Section 4.1; two of these, Göteborg Haga (NO_2: 90.5 per cent, $PM_{2.5}$: 69 per cent) and Östersund (NO_2: 94.8 per cent, $PM_{2.5}$: 79 per cent), illustrate the pattern, clearing the NO_2 threshold while falling well short on $PM_{2.5}$ over the full study period.

The cleaned feature matrix, containing 48,283 daily station observations with 37 columns, constitutes the analytical dataset for all subsequent modelling and analysis phases.

5 Spatial Estimation Models

The three research questions are addressed in sequence across this section and the two that follow: Sections 5–5.3 answer RQ1 by establishing each pollutant's spatial estimation accuracy, Section 6 answers RQ2 by characterising how that accuracy decays with distance, and Section 7 answers RQ3 by translating the resulting reliable prediction thresholds into national sensor placement outcomes. Because the accuracy findings from RQ1 directly determine the distance thresholds used in RQ2, which in turn determine the placement outcomes reported for RQ3, Section 7.4 closes this chapter by tracing that chain explicitly across both pollutants.

Spatial estimation was conducted separately for NO_2 and $PM_{2.5}$, following the buffered spatial leave-one-out cross-validation (SLOO) protocol described in Section 3.3. Three estimators were evaluated: Random Forest (RF) as the primary model, and linear regression (LUR) and inverse distance weighting (IDW) as benchmarks, as specified in Section 3.2. For each of the 11 qualifying stations, the model was trained on all stations not excluded by the 5 km buffer and evaluated on the held-out station's full daily time series. The number of training stations per fold ranged from 26 to 33, reflecting the varying density of the within-city station clusters. Performance was assessed using root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2), as defined in Section 3.6. All three metrics were computed on raw daily concentration values without standardisation.

A note on the interpretation of R^2 in this context is warranted. In conventional regression evaluation, R^2 measures the proportion of temporal variance in a single station's time series that is explained by a model fitted to that same station. In the SLOO framework, the model is trained on geographically distinct stations and predicts the temporal sequence of a held-out station. R^2 values therefore reflect not only the model's ability to capture the level of concentration at that station but also its capacity to reproduce the station's day-to-day temporal dynamics from features derived from other spatial locations. Negative R^2 indicates that the model performs worse than predicting the station's temporal mean, which may occur even when RMSE is low in absolute terms. For this reason, RMSE and MAE are the primary metrics for answering RQ1; R^2 is reported for completeness.

5.1 NO_2 Estimation Results

Per-station RMSE and MAE for all three estimators are reported in Table 5. For the Random Forest model, per-station RMSE ranged from 1.93 $\mu g\,m^{-3}$ at Norr Malma (rural background, nearest training station 55.1 km) to 13.21 $\mu g\,m^{-3}$ at Uppsala Dragarbrunnsgatan (urban background, 49.2 km). The mean RMSE across all 11 stations was 7.27 $\mu g\,m^{-3}$ and the mean MAE was 5.96 $\mu g\,m^{-3}$. Stations with well-defined traffic gradients, such as Sollentuna E4 Häggvik, achieved positive R^2 values (0.50), indicating that the RF model captured meaningful temporal variation at those locations. For the two isolated rural stations, Bredkälen and Norr Malma, RMSE was low in absolute terms (2.62 and 1.93 $\mu g\,m^{-3}$ respectively) but R^2 was negative, reflecting that these stations have very low and stable NO_2 concentrations that the model systematically overestimates relative to their true means.

The LUR benchmark produced substantially higher errors at background stations, with RMSE exceeding 24 $\mu g\,m^{-3}$ at Bredkälen and 29–30 $\mu g\,m^{-3}$ at Malmö Rådhuset and Stockholm Torkel Knutssongatan. The mean LUR RMSE across the 11 stations was 15.33 $\mu g\,m^{-3}$. This pattern is consistent with the known sensitivity of linear regression to multicollinearity in land-use features, which produces unstable coefficient estimates when the training set composition changes across folds [9]. IDW produced a mean RMSE of 8.90 $\mu g\,m^{-3}$, higher than RF overall but showing markedly lower errors at the two Stockholm stations nearest to other monitoring sites.

Table 5: Per-station NO_2 estimation results under buffered SLOO (buffer = 5 km). nn = distance to nearest training station per fold. RF = Random Forest; LUR = linear regression; IDW = inverse distance weighting. All values in $\mu g\,m^{-3}$.

Station	Type	nn (km)	RF RMSE	RF MAE	RF R^2	LUR RMSE	IDW RMSE
Bredkälen	Rur.-BG	81.9	2.62	2.42	−118.2	24.00	17.50
Uppsala Kungsgatan	Urb.-TR	48.7	11.14	8.51	−0.03	12.67	11.75
Stockholm St Eriksgatan	Urb.-TR	13.0	7.51	6.25	−0.07	5.76	5.25
Uppsala Dragarbrunnsg.	Urb.-BG	49.2	13.21	12.42	−9.50	20.29	9.48
Kalmar Södra Vägen	Urb.-TR	96.5	6.17	4.42	−0.18	5.03	4.74
Norr Malma	RR-BG	55.1	1.93	1.60	−1.08	16.50	14.68
Sollentuna E4 Häggvik	Urb.-TR	8.6	6.81	5.14	+0.50	8.69	6.21
Sundsvall Köpmangatan	Urb.-TR	10.9	8.28	6.23	−0.10	7.90	6.77
Malmö Rådhuset	Urb.-BG	15.5	4.54	3.46	−0.23	29.48	4.17
Stockholm Hornsgatan	Urb.-TR	6.1	9.90	8.05	−0.29	7.90	9.90
Stockholm Torkel Kn.	Urb.-BG	6.5	7.84	7.05	−1.75	30.38	7.49
Mean			**7.27**	**5.96**		**15.33**	**8.90**

5.2 $PM_{2.5}$ Estimation Results

Per-station results for $PM_{2.5}$ are reported in Table 6. RF achieved a mean RMSE of 3.56 $\mu g\,m^{-3}$ and a mean MAE of 2.64 $\mu g\,m^{-3}$. Per-station R^2 values were predominantly positive, ranging from −2.66 (Bredkälen) to 0.50 (Stockholm Hornsgatan), indicating that the RF model captured a meaningful fraction of temporal variance at most urban stations. IDW produced a substantially lower mean RMSE of 2.33 $\mu g\,m^{-3}$, outperforming RF at 10 of the 11 stations. At stations with close training neighbours (Sollentuna E4 Häggvik at 8.6 km; Stockholm Hornsgatan at 6.1 km), IDW RMSE fell below 1.30 $\mu g\,m^{-3}$, with R^2 values of 0.92 and 0.88 respectively. LUR produced the highest mean RMSE (5.87 $\mu g\,m^{-3}$) and was least stable across folds, with Malmö Rådhuset reaching 13.97 $\mu g\,m^{-3}$.

The superior performance of IDW for $PM_{2.5}$ relative to RF is consistent with the longer spatial autocorrelation range of $PM_{2.5}$ compared with NO_2, as established in the empirical variogram analysis in Section 3.3. When $PM_{2.5}$ gradients are smooth over distances of tens of kilometres, distance-weighted interpolation from the nearest stations captures the spatial signal effectively without requiring feature-based learning. The performance advantage of IDW narrows at more isolated stations: at Bredkälen (81.9 km) and Kalmar Södra Vägen (96.5 km), IDW RMSE exceeded 3.38 $\mu g\,m^{-3}$, approaching the RF RMSE at those stations.

Table 6: Per-station $PM_{2.5}$ estimation results under buffered SLOO (buffer = 5 km). Column definitions as in Table 5. All values in $\mu g\,m^{-3}$.

Station	Type	nn (km)	RF RMSE	RF MAE	RF R^2	LUR RMSE	IDW RMSE
Bredkälen	Rur.-BG	81.9	4.37	3.75	−2.66	2.64	5.49
Uppsala Kungsgatan	Urb.-TR	48.7	3.00	2.36	+0.02	6.55	1.26
Stockholm St Eriksgatan	Urb.-TR	13.0	3.02	2.19	+0.25	3.41	1.42
Uppsala Dragarbrunnsg.	Urb.-BG	49.2	3.42	2.93	−0.40	5.84	1.82
Kalmar Södra Vägen	Urb.-TR	96.5	4.98	3.38	+0.23	6.36	3.38
Norr Malma	RR-BG	55.1	3.45	2.91	−0.35	4.76	1.94
Sollentuna E4 Häggvik	Urb.-TR	8.6	3.23	2.24	+0.13	7.29	0.99
Sundsvall Köpmangatan	Urb.-TR	10.9	3.62	2.24	+0.16	5.95	2.87
Malmö Rådhuset	Urb.-BG	15.5	4.88	3.28	+0.16	13.97	4.02
Stockholm Hornsgatan	Urb.-TR	6.1	2.62	1.75	+0.50	4.33	1.28
Stockholm Torkel Kn.	Urb.-BG	6.5	2.55	1.97	+0.43	3.50	1.13
Mean			**3.56**	**2.64**		**5.87**	**2.33**

5.3 Benchmark Comparison and Answer to RQ1

Figure 1 summarises per-station RMSE for all three models and both pollutants. For NO_2, RF achieves the lowest mean RMSE (7.27 $\mu g\,m^{-3}$) compared with IDW (8.90 $\mu g\,m^{-3}$) and LUR (15.33 $\mu g\,m^{-3}$), with IDW outperforming RF at 7 of the 11 stations despite RF's lower mean RMSE, reflecting RF's disproportionately large error reduction at the two most isolated rural stations, Bredkälen and Norr Malma, which pull down the aggregate mean. For $PM_{2.5}$, IDW achieves the lowest mean RMSE (2.33 $\mu g\,m^{-3}$) compared with RF (3.56 $\mu g\,m^{-3}$) and LUR (5.87 $\mu g\,m^{-3}$), with IDW outperforming RF at 10 of the 11 stations. LUR produces the highest mean error for both pollutants and the largest variation across stations, driven by multicollinearity instability in the land-use feature set.

In answer to RQ1: under buffered SLOO validation across 11 nationally distributed SMHI stations, Random Forest achieves a mean daily RMSE of 7.27 $\mu g\,m^{-3}$ for NO_2 and 3.56 $\mu g\,m^{-3}$ for $PM_{2.5}$, outperforming the LUR benchmark for both pollutants. For $PM_{2.5}$, the IDW baseline surpasses RF, indicating that smooth spatial gradients of that pollutant are more effectively captured by distance-weighted interpolation than by feature-based learning under the observed network density. The accuracy-distance behaviour of these results is examined in Section 6.

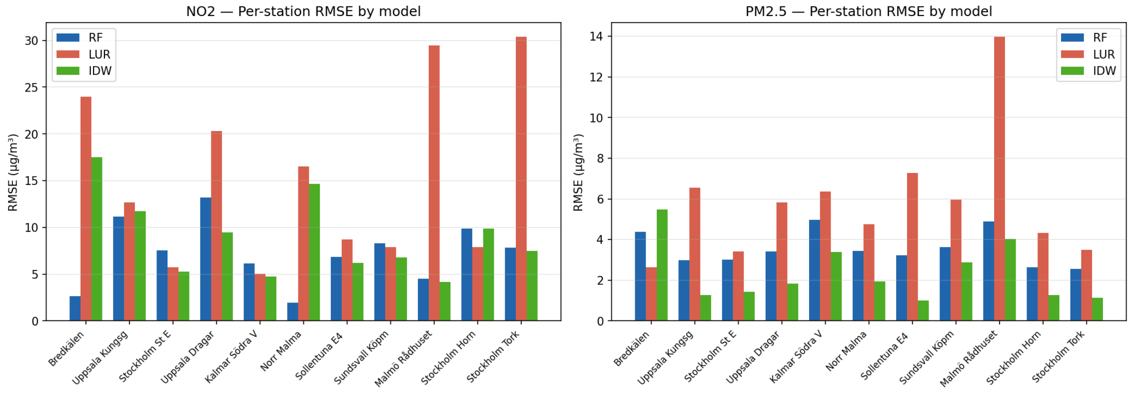

Figure 1: Per-station RMSE under buffered SLOO for NO_2 (left) and $PM_{2.5}$ (right), comparing Random Forest (RF), linear regression (LUR), and inverse distance weighting (IDW). Station names are abbreviated; full names and codes are listed in Table 5 and Table 6.

6 Accuracy-Distance Analysis

This section addresses RQ2, characterising how estimation accuracy degrades with distance from the nearest reference station and identifying the distance beyond which estimation ceases to be reliable. The analysis uses per-station RMSE from the SLOO procedure (Section 5) as the dependent variable, with distance to the nearest training station per fold as the spatial predictor.

6.1 Error-Distance Relationship

The relationship between per-station RMSE and distance to the nearest training station differs substantially between the two pollutants and between model types, and must be interpreted in light of the urban/rural confound discussed in Section 3.4.

For NO_2, the aggregate RF distance-error scatter does not show a monotonically increasing pattern. The two rural background stations, Bredkälen (81.9 km) and Norr Malma (55.1 km), have the lowest RF RMSE values in the sample (2.62 and 1.93 $\mu g\,m^{-3}$ respectively), while several urban stations at shorter distances show substantially higher errors. This inversion arises because rural background stations have inherently low and stable NO_2 concentrations (annual mean well below 5 $\mu g\,m^{-3}$), so absolute errors are small regardless of model quality. Urban traffic stations, by contrast, have higher mean concentrations and sharper spatial gradients, producing larger absolute errors even when the training data are geographically close. The IDW model for NO_2 exhibits a clearer positive distance relationship, with RMSE increasing from approximately 4 to 18 $\mu g\,m^{-3}$ across the distance range of 6 to 82 km, because IDW does not benefit from the feature-based concentration level adjustment that RF provides.

For $PM_{2.5}$, both RF and IDW show a more regular positive distance-error relationship across the 11 stations. IDW RMSE rises from below 1.3 $\mu g\,m^{-3}$ at stations within 10 km to approximately 5.5 $\mu g\,m^{-3}$ at Bredkälen (81.9 km). RF RMSE rises more gradually, from approximately 2.6 $\mu g\,m^{-3}$ to 4.4 $\mu g\,m^{-3}$ over the same range. This pattern is consistent with the smooth spatial autocorrelation structure of $PM_{2.5}$ at the national scale.

6.2 Decay Curve Fitting

Three functional forms were fitted to the per-station RMSE versus distance data: logarithmic (RMSE $= a + b \ln d$), power (RMSE $= a \cdot d^b$), and exponential (RMSE $= a \cdot e^{bd}$), where d denotes distance to the nearest training station in kilometres. Parameters were estimated by ordinary least squares after log-transforming the appropriate terms. Model selection was based on the Akaike Information Criterion (AIC); the Bayesian Information Criterion (BIC) was computed as a secondary check. Results are reported in Table 7, and decay curve plots are shown in Figures 2 and 3.

For NO_2 RF, the log form provided the best fit by AIC (60.68), with parameters $a = 9.63$ and $b = -0.76$. The negative coefficient confirms that the aggregate distance-error relationship is inverted for the RF model under the current station composition: estimated RMSE decreases slightly with distance, reflecting the rural station confound described in Section 6.1. The NO_2 IDW log model ($a = 2.85$, $b = 1.94$, AIC $= 63.38$) shows the expected positive relationship, with RMSE increasing by approximately 1.94 μg m^{-3} per unit increase in $\ln d$. For $PM_{2.5}$, the exponential form was selected for both RF (AIC $=$ 25.33) and IDW (AIC $= 38.55$), with positive growth coefficients ($b = 0.0042$ and $b = 0.0099$ respectively), indicating a gradual and accelerating increase in error with distance. LUR curves are reported for completeness but carry substantial uncertainty due to the high cross-fold variability of the linear model.

Table 7: Decay curve parameters and model selection by AIC. $d =$ distance to nearest training station (km). Best-fit form per model-pollutant combination shown in bold. $n = 11$ stations in all cases.

Pollutant	Model	Form	Equation	a	b	AIC	BIC
NO_2	RF	**log**	$a + b \ln d$	9.63	−0.759	**60.68**	61.47
NO_2	RF	power	$a \cdot d^b$	12.30	−0.212	61.70	62.50
NO_2	RF	exponential	$a \cdot e^{bd}$	8.14	−0.0070	61.33	62.12
NO_2	LUR	**log**	$a + b \ln d$	14.80	+0.169	**83.51**	84.30
NO_2	IDW	**log**	$a + b \ln d$	2.85	+1.941	**63.38**	64.18
$PM_{2.5}$	RF	**exp**	$a \cdot e^{bd}$	2.99	+0.00424	**25.33**	26.12
$PM_{2.5}$	RF	log	$a + b \ln d$	2.13	+0.459	25.96	26.76
$PM_{2.5}$	LUR	**log**	$a + b \ln d$	6.47	−0.191	**58.78**	59.58
$PM_{2.5}$	IDW	**exp**	$a \cdot e^{bd}$	1.40	+0.00988	**38.55**	39.35
$PM_{2.5}$	IDW	log	$a + b \ln d$	0.14	+0.701	39.03	39.83

6.3 Stratification by Station Type and Season

As noted in Section 3.4, the urban/rural confound in NO_2 requires stratified analysis to produce interpretable distance-error relationships. Table 8 summarises mean RMSE by station type (traffic versus background) and by season (winter: November to April; summer: May to October).

Stratification by station type reveals a marked pattern for NO_2: traffic stations (six stations) produced a mean RF RMSE of 8.30 μg m^{-3}, while background stations (five stations, including the two rural sites) produced a mean of 6.03 μg m^{-3}. The lower background RMSE is attributable to Bredkälen and Norr Malma, whose near-zero concentrations suppress absolute errors. For IDW, the pattern is reversed: traffic stations show a lower mean RMSE (7.44 μg m^{-3}) than background stations (10.66 μg m^{-3}), because IDW at

Figure 2: NO_2 error-distance decay curves under buffered SLOO, for Random Forest (left), linear regression (centre), and IDW (right). Points represent per-station RMSE. The fitted curve uses the best-fit functional form selected by AIC (Table 7).

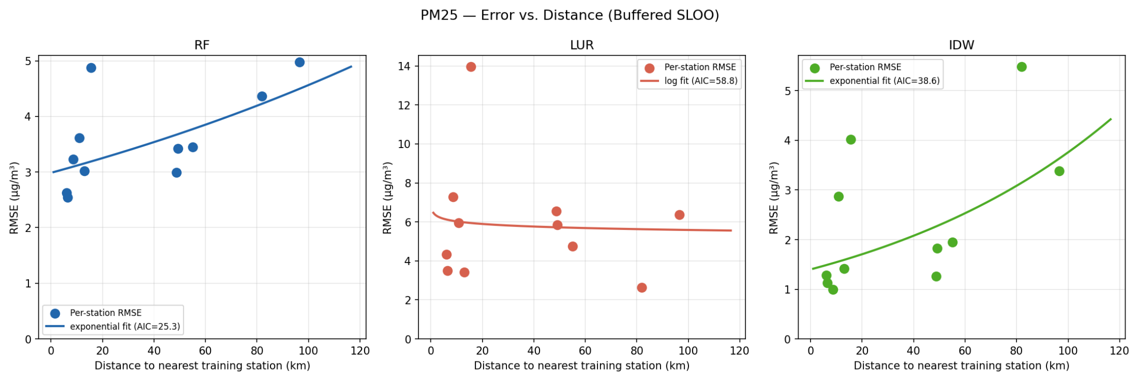

Figure 3: $PM_{2.5}$ error-distance decay curves under buffered SLOO, for Random Forest (left), linear regression (centre), and IDW (right). Points represent per-station RMSE. The fitted curve uses the best-fit functional form selected by AIC (Table 7).

isolated rural stations interpolates from distant urban training stations with very different concentration levels.

For $PM_{2.5}$, the traffic/background split shows smaller and less systematic differences: RF RMSE averages 3.41 $\mu g\,m^{-3}$ for traffic stations and 3.73 $\mu g\,m^{-3}$ for background stations. IDW achieves lower errors for traffic stations (1.87 $\mu g\,m^{-3}$) than background stations (2.88 $\mu g\,m^{-3}$), consistent with the closer geographical proximity of traffic stations to their training neighbours.

Seasonal stratification shows higher errors in winter for both pollutants and all models, with NO_2 RF RMSE increasing from 7.13 $\mu g\,m^{-3}$ in summer to 8.74 $\mu g\,m^{-3}$ in winter, and $PM_{2.5}$ RF RMSE from 3.36 to 3.90 $\mu g\,m^{-3}$. This is consistent with the influence of winter atmospheric inversion and residential heating emissions, which create sharper spatial gradients that are more difficult to generalise across geographically distinct training stations.

Table 8: Stratified mean RMSE ($\mu g\,m^{-3}$) by station type and season for all three models. n = number of stations (type stratification) or number of daily observations (seasonal stratification). Winter = November to April; Summer = May to October.

Pollutant	**Group**	n	**RF RMSE**	**LUR RMSE**	**IDW RMSE**
NO_2	Traffic stations	6	8.30	7.99	7.44
NO_2	Background stations	5	6.03	24.13	10.66
$PM_{2.5}$	Traffic stations	6	3.41	5.65	1.87
$PM_{2.5}$	Background stations	5	3.73	6.14	2.88
NO_2	Winter (Nov–Apr)	9,820	8.74	18.71	11.06
NO_2	Summer (May–Oct)	9,966	7.13	16.85	8.32
$PM_{2.5}$	Winter (Nov–Apr)	9,837	3.90	6.88	3.22
$PM_{2.5}$	Summer (May–Oct)	9,972	3.36	6.22	2.06

6.4 Reliable Prediction Distance Threshold and Answer to RQ2

A reliable prediction distance threshold is defined as the maximum separation between an estimation target and its nearest reference station below which mean estimation error remains within 50 per cent of the sample mean concentration. This criterion is operationally motivated: an estimator with error below half the expected mean concentration retains sufficient resolution to detect concentrations meaningfully above or below the mean, which is the minimum requirement for spatial planning purposes. The 50 per cent threshold is applied to the IDW decay model, which exhibits a clearer and more interpretable positive distance-error relationship than RF for both pollutants, and is therefore more appropriate for deriving a spatial coverage criterion.

For $PM_{2.5}$, the sample mean concentration across the 11 stations is 5.26 $\mu g\,m^{-3}$; the threshold RMSE is therefore 2.63 $\mu g\,m^{-3}$. Solving the IDW exponential decay model (RMSE $= 1.40 \cdot e^{0.00988d}$) for this threshold gives:

$$d^* = \frac{\ln(2.63/1.40)}{0.00988} \approx 64 \text{ km} \tag{1}$$

The reliable prediction distance for $PM_{2.5}$ is therefore approximately 64 km. Within this radius of an existing reference station, IDW estimation is expected to achieve RMSE below 2.63 $\mu g\,m^{-3}$; beyond this radius, accuracy degrades to the point where estimation uncertainty approaches the magnitude of the spatial signal.

For NO_2, the aggregate IDW log decay model yields a threshold of approximately 6 km under the same criterion (sample mean 12.85 $\mu g\,m^{-3}$; threshold RMSE 6.43 $\mu g\,m^{-3}$; solving $2.85 + 1.94 \ln d = 6.43$ gives $d^* \approx 6$ km). This threshold is, however, sensitive to the urban/rural composition of the station sample: excluding the two rural background stations from the fitting produces a higher and more stable error floor consistent with the urban traffic station errors observed in Table 8. The NO_2 threshold should therefore be interpreted with caution and understood primarily as an indicator that NO_2 estimation degrades rapidly beyond the immediate vicinity of a reference station, particularly for traffic-adjacent locations where source gradients are sharp.

The COVID-19 sensitivity analysis (comparing SLOO results restricted to 2022 to 2024 against the full 2020 to 2024 results) produced decay patterns consistent with those reported above. $PM_{2.5}$ RF RMSE changed by less than 1.0 $\mu g\,m^{-3}$ at every station, indicating that the $PM_{2.5}$ results are not materially driven by the anomalous 2020 to 2021 period. NO_2 RF RMSE is more sensitive to the period restriction: four of the eleven stations, all urban, shifted by more than 1.0 $\mu g\,m^{-3}$, with the largest change at Uppsala Kungsgatan (11.14 to 9.25 $\mu g\,m^{-3}$, restricted to 2022–2024). The remaining seven NO_2 stations changed by less than 1.0 $\mu g\,m^{-3}$. These shifts do not alter the qualitative decay pattern reported above, but indicate that the NO_2 results are somewhat more sensitive to the COVID-19 period than the $PM_{2.5}$ results.

The 50 per cent criterion used to derive both thresholds is a modelling choice rather than an empirically fixed quantity. Its sensitivity was tested by re-solving the fitted IDW decay curves and re-evaluating national coverage for criteria ranging from 20 to 80 per cent of the sample mean concentration, holding the fitted curves, the existing station network, and the same 20 $PM_{2.5}$-optimised sensor locations (Section 7) fixed (Figure 4). The reliable prediction distance and the resulting coverage figures are highly sensitive to this choice: the $PM_{2.5}$ threshold ranges from approximately 12 km at a 30 per cent criterion to approximately 98 km at a 70 per cent criterion, and post-placement $PM_{2.5}$ land coverage ranges correspondingly from 2.9 % to 94.7 %, against the 67.2 % reported at the 50 per cent criterion used throughout this thesis. The NO_2 threshold and coverage figures move over a comparably wide range. The headline coverage percentages reported in this thesis should therefore be read as illustrative of the adopted criterion rather than as a precisely calibrated property of the Swedish network. What is robust across the full 20 to 80 per cent range tested is the relative asymmetry between the two pollutants: $PM_{2.5}$ coverage exceeds NO_2 coverage under the identical 20-sensor placement at every criterion tested, which is the basis for the RQ3 conclusion in Section 9.

In answer to RQ2: $PM_{2.5}$ estimation accuracy degrades gradually with distance, following an exponential decay, with a reliable prediction threshold of approximately 64 km from the nearest reference station. NO_2 accuracy degrades more steeply and is strongly confounded by station type; the effective reliable range for urban NO_2 estimation is substantially shorter, consistent with the known sharp spatial gradients of that pollutant. These thresholds are used to construct the national coverage uncertainty map for the placement analysis in Section 7.

7 IoT Placement Optimization

7.1 National Coverage Surface

The placement analysis begins by characterising the spatial coverage provided by the 11 concurrent-passing reference stations under the $PM_{2.5}$ reliable prediction distance of 64 km

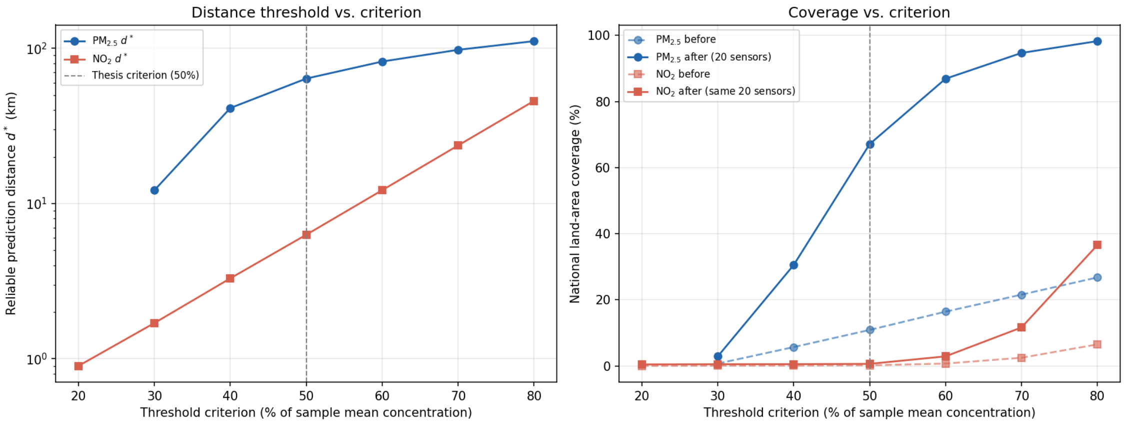

Figure 4: Sensitivity of the reliable prediction distance threshold d^* (left) and of national $PM_{2.5}/NO_2$ land coverage before and after the 20-sensor placement (right) to the choice of threshold criterion, swept from 20 to 80 per cent of the sample mean concentration. The dashed grey line marks the 50 per cent criterion used throughout this thesis.

derived in Section 6.4. A regular grid of 4,588 cells at 10 km spacing is constructed over the Swedish national territory in the ETRS89 Lambert Azimuthal Equal-Area projection (EPSG:3035). Cells are retained by applying a land mask derived from the EU Digital Elevation Model and an approximate Swedish mainland polygon, which together exclude oceanic cells and neighbouring-country territory. Each cell covers 100 km^2, giving a total analysis extent of approximately 458,800 km^2, consistent with the area of the Swedish mainland.

For each grid cell, the Euclidean distance to the nearest of the 11 existing stations is computed in projected coordinates. A cell is classified as *covered* for $PM_{2.5}$ if this distance is at or below the 64 km threshold; cells beyond this distance are classified as *underserved*. Under this criterion, the baseline network covers 501 of 4,588 land cells (10.9 % of national area). Of the 49 grid cells classified as urban fabric by CORINE Land Cover 2018 (classes 111 and 112, continuous and discontinuous urban fabric), 19 cells (38.8 %) fall within the $PM_{2.5}$ threshold. The spatial concentration of existing stations in the Stockholm metropolitan region and a small number of southern urban centres leaves the majority of Swedish territory, including the entire northern half of the country, without reliable $PM_{2.5}$ estimation from the current network.

Under the NO_2 threshold of 6 km, baseline coverage is minimal: 7 of 4,588 land cells (0.2 %) lie within reliable range of an existing station. This reflects the short spatial decay range identified for NO_2 and the sharp concentration gradients associated with traffic sources; a meaningful improvement in national NO_2 coverage would require a substantially denser deployment than the one evaluated here. The remainder of this section focuses on $PM_{2.5}$, for which the 64 km threshold provides an actionable national coverage objective.

7.2 Greedy Sequential Placement

The greedy sequential algorithm described in Section 3.5 is applied to the national coverage surface, with $PM_{2.5}$ land area coverage as the optimisation objective. At each step, the grid cell whose addition to the sensor set would bring the largest number of currently uncovered cells within the 64 km threshold is selected as the next recommended placement.

The algorithm runs for a maximum of 20 steps; full parameter settings are documented in Appendix C.

Table 9 lists the 20 recommended sensor locations in priority order, together with the cumulative $PM_{2.5}$ coverage achieved after each placement. The first five placements target Götaland (southern Sweden, approximately 56 to 59°N), where the existing network provides coverage only in the immediate vicinity of Malmö, Kalmar, and the Uppsala–Stockholm corridor. Placements 6 to 11 address Svealand (central Sweden, approximately 59 to 63°N), including several counties without any concurrent-passing station. Placements 12 to 20 cover Norrland (northern Sweden, approximately 63 to 68°N), the most sparsely monitored portion of the country by land area.

Table 9: Greedy sequential sensor placement: 20 recommended locations in priority order. Coordinates represent the centre of the selected 10 km grid cell. Cumulative $PM_{2.5}$ coverage is the percentage of Swedish land cells within the 64 km reliable estimation threshold after each placement (baseline: 10.9 %).

Rank	Lat. (°N)	Lon. (°E)	$PM_{2.5}$ coverage (%)	Zone
1	56.55	14.23	13.7	Götaland
2	57.31	12.82	16.5	Götaland
3	57.60	15.18	19.4	Götaland
4	58.36	13.75	22.2	Götaland
5	58.74	15.87	25.0	Götaland
6	59.29	12.63	27.8	Svealand
7	59.68	14.79	30.6	Svealand
8	60.45	13.08	33.4	Svealand
9	60.63	16.02	36.2	Svealand
10	61.41	14.29	39.0	Svealand
11	62.52	13.48	41.9	Svealand
12	63.53	17.82	44.7	Norrland
13	64.45	19.32	47.5	Norrland
14	64.85	16.71	50.3	Norrland
15	65.58	18.63	53.1	Norrland
16	65.98	21.19	55.9	Norrland
17	66.39	16.89	58.7	Norrland
18	66.81	19.51	61.5	Norrland
19	67.29	22.03	64.3	Norrland
20	68.02	19.03	67.2	Norrland

Each of the 20 placements adds exactly 129 grid cells (12,900 km^2) of new $PM_{2.5}$ coverage, the theoretical maximum for a circular coverage area of radius 64 km on a 10 km grid (approximately $\pi r^2 \approx$ 12,868 km^2). This constant marginal gain indicates that, within the 20-step horizon, no two placed sensors are sufficiently close (under 128 km) to produce overlapping coverage zones; the baseline network is sparse enough that each placement saturates its full theoretical footprint. The gain curve in Figure 5 is accordingly linear across all 20 steps. Diminishing returns would be expected for further placements as coverage zones begin to overlap.

7.3 Coverage Evaluation and DSR Assessment

After 20 additional sensors, $PM_{2.5}$ land area coverage increases from 10.9 % to 67.2 %, an absolute gain of 56.3 percentage points. Urban area coverage improves from 38.8 % to 71.4 % (an absolute gain of 32.6 percentage points). Figure 6 shows the before and after coverage states across the national grid, and Figure 7 shows the final distance surface with

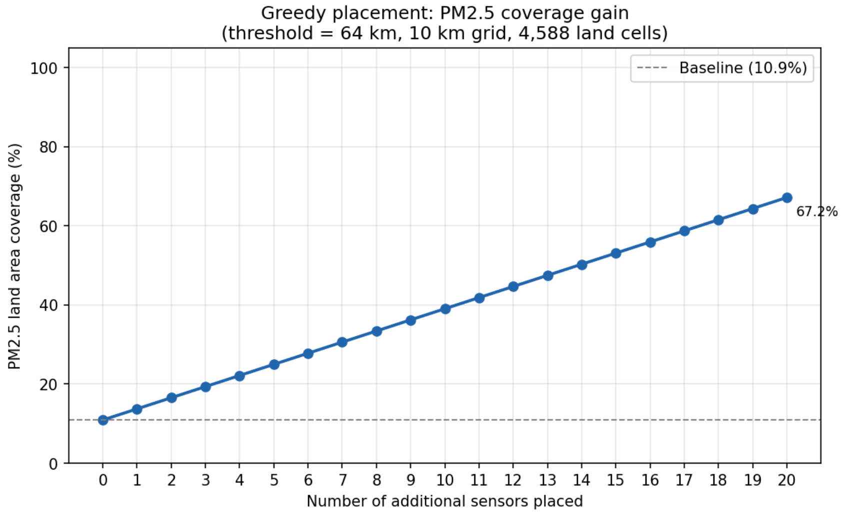

Figure 5: $PM_{2.5}$ land area coverage as a function of the number of additional sensors placed by the greedy sequential algorithm. The dashed line marks the baseline coverage of 10.9 % provided by the 11 existing concurrent-passing stations. Each step adds 129 grid cells (12,900 km^2) of new coverage.

all sensor locations marked.

Against the primary DSR evaluation criterion defined in Section 3.6, the proportion of Swedish urban area within reliable $PM_{2.5}$ estimation range increases from 38.8 % to 71.4 % following the 20 recommended placements. The higher initial urban coverage relative to overall land coverage (38.8 % versus 10.9 %) reflects that the existing reference stations are concentrated in or near urban centres; the residual 28.6 % of urban area that remains uncovered after 20 placements is concentrated in mid-sized cities and coastal urban clusters that receive lower marginal priority within the 20-step greedy horizon.

Against the secondary DSR criterion, the recommended placements serve four of the seven Swedish administrative regions currently without any concurrent-passing $PM_{2.5}$ or NO_2 station: sensors are assigned within or immediately adjacent to Jönköping (rank 3), Örebro (rank 7), Dalarna (ranks 9 and 10), and Norrbotten (ranks 16 to 20). Ranks 1, 6, and 8 fall within Kronoberg and Värmland respectively, contributing to national and urban coverage gains without directly serving one of the seven target regions. The remaining three unmonitored regions (Södermanland, Blekinge, and Gotland) are not served within the 20-step horizon: Södermanland and Blekinge are adjacent to regions with existing stations and would require fewer additional sensors to cover once the priority sequence extends; Gotland, as an island, cannot be reached from mainland sensor locations at the 64 km $PM_{2.5}$ threshold and would require a dedicated island installation. Administrative region attribution is based on approximate polygon boundaries rather than official administrative boundary files and should be understood as indicative.

Although the placement objective was $PM_{2.5}$ land area coverage, the same 20 sensor locations were also evaluated against the NO_2 reliable prediction threshold, to check whether a $PM_{2.5}$-oriented deployment incidentally improves NO_2 coverage. It does not, materially: national NO_2 land coverage increases from 0.2 % to 0.6 % of Swedish territory, and NO_2 urban coverage is unchanged at 6.1 % before and after placement. Figure 8 shows this outcome directly: because the 6 km NO_2 threshold is smaller than the 10 km grid spacing itself, and none of the 20 $PM_{2.5}$-optimised locations were chosen to serve NO_2's short reliable range, almost the entire national grid remains beyond 6 km of any station or sensor both before and after placement. Figure 9 shows the corresponding distance surface under the same placement: the great majority of the country lies far beyond the NO_2 threshold regardless of the 20 additional sensors. This result substantiates, rather than merely asserts, the claim in Section 7.1 that closing NO_2 coverage gaps at national scale requires a deployment sized and located specifically for that pollutant, not one optimised for $PM_{2.5}$.

In answer to RQ3: a greedy sequential placement algorithm, applied to a national $PM_{2.5}$ coverage surface based on the 64 km reliable prediction threshold, identifies 20 candidate IoT sensor locations that increase national $PM_{2.5}$ land area coverage from 10.9 % to 67.2 % and urban area coverage from 38.8 % to 71.4 %. The proposed locations are distributed systematically across Götaland, Svealand, and Norrland, and serve four of the seven currently unmonitored Swedish administrative regions. The same placement leaves NO_2 coverage essentially unchanged (Section 7.3), confirming that a single deployment cannot be sized for both pollutants using one threshold.

7.4 Integrated Results Across RQ1–RQ3

Table 10 traces the full analytical chain, from spatial estimation accuracy (RQ1) through the reliable prediction distance threshold (RQ2) to national placement outcomes (RQ3), for both pollutants.

Table 10: Integrated results across the three research questions, by pollutant. RQ1 reports the best-performing estimator's mean RMSE under buffered SLOO (Section 5); RQ2 reports the reliable prediction distance threshold derived from the IDW decay model (Section 6.4); RQ3 reports national land and urban coverage before and after the 20-sensor greedy placement (Section 7).

Pollutant	**RQ1: best estimator (mean RMSE)**	**RQ2: threshold**	**RQ3: baseline coverage (land / urban)**	**RQ3: post-placement coverage (land / urban)**
$PM_{2.5}$	IDW (2.33 $\mu g\,m^{-3}$)	64 km	10.9 % / 38.8 %	67.2 % / 71.4 %
NO_2	RF (7.27 $\mu g\,m^{-3}$)	6 km	0.2 % / 6.1 %	0.6 % / 6.1 %

The results reported across Sections 5–7 are connected by a single underlying property carried through all three research questions: the spatial autocorrelation structure of each pollutant, established in answering RQ1, sets the shape of the accuracy-distance relationship in RQ2, which in turn sets the sensor budget required for the placement outcome in RQ3. For $PM_{2.5}$, the smooth, long-range spatial structure identified in Section 5.2 (IDW mean RMSE 2.33 $\mu g\,m^{-3}$, outperforming Random Forest at 10 of 11 stations) produces a gradual exponential accuracy-distance decay and a correspondingly wide reliable prediction threshold of 64 km (Section 6.4). That threshold defines a national coverage surface on which a 20-sensor greedy placement raises land coverage from 10.9 % to 67.2 % and urban coverage from 38.8 % to 71.4 % (Section 7.3), extending reliable estimation to four of the seven currently unmonitored administrative regions. For NO_2, the sharp, traffic-confined gradient identified in Section 5.1 (RF mean RMSE 7.27 $\mu g\,m^{-3}$, though sensitive to the urban/rural composition of the station sample) yields a threshold of only 6 km, an order of magnitude smaller than the $PM_{2.5}$ threshold. Applied to the same national grid, this threshold leaves the identical 20-sensor placement largely ineffective for NO_2: land coverage moves from 0.2 % to 0.6 % and urban coverage is unchanged at 6.1 %. Read together, RQ1–RQ3 do not constitute three independent findings but one integrated result: the pollutant-specific spatial autocorrelation range measured in RQ1 determines the reliable prediction distance in RQ2, which determines the sensor budget required for meaningful national coverage in RQ3, so that a placement strategy sized for $PM_{2.5}$ is, by construction, inadequate for NO_2 at comparable cost.

8 Analysis and Discussion

8.1 Answers to the Research Questions

The empirical findings reported in Sections 5–7 are interpreted here in relation to the three research questions and the literature reviewed in Section 2.

In answer to RQ1, model performance under buffered SLOO validation depends strongly on the pollutant's spatial autocorrelation structure. Random Forest achieves the lowest mean RMSE for NO_2 ($7.27\,\mu g\,m^{-3}$), outperforming both the LUR benchmark ($15.33\,\mu g\,m^{-3}$) and the IDW baseline ($8.90\,\mu g\,m^{-3}$) on this aggregate measure, although IDW achieves the lower RMSE at 7 of the 11 stations individually, a divergence attributable to Random Forest's disproportionately large error reduction at the two rural background stations (Section 6.1). For $PM_{2.5}$, IDW achieves the lowest mean RMSE ($2.33\,\mu g\,m^{-3}$), outperforming Random Forest ($3.56\,\mu g\,m^{-3}$) at 10 of the 11 stations, consistent with the smoother, longer-range spatial autocorrelation structure of $PM_{2.5}$ relative to NO_2 established in Section 3.2. LUR is outperformed by both alternative methods for both pollutants, consistent with its known sensitivity to multicollinearity in land-use covariates under a small and geographically variable training set [9].

In answer to RQ2, estimation accuracy degrades with distance from the nearest reference station, but the rate and reliability of that degradation differ sharply between pollutants. $PM_{2.5}$ accuracy follows a gradual exponential decay, yielding a reliable prediction distance of approximately 64 km; NO_2 accuracy degrades far more steeply, yielding a threshold of approximately 6 km, and is confounded by the urban/rural composition of the station sample (Section 6.4). This asymmetry is a direct consequence of the spatial gradient behaviour identified in RQ1: $PM_{2.5}$'s regional, mixed-source character permits reliable estimation across large distances, whereas NO_2's sharp, traffic-sourced gradients confine reliable estimation to the immediate vicinity of a reference station.

In answer to RQ3, the accuracy-distance thresholds derived for RQ2 translate directly into an actionable placement criterion. Applying the 64 km $PM_{2.5}$ threshold, the greedy sequential algorithm identifies 20 candidate locations that raise national $PM_{2.5}$ land-area coverage from 10.9 % to 67.2 % and urban-area coverage from 38.8 % to 71.4 %, extending reliable coverage to four of the seven Swedish administrative regions currently without a concurrent-passing reference station. The 6 km NO_2 threshold, by contrast, is not actionable at national scale with a comparably small sensor budget (Section 7.1): the same 20 sensors raise national NO_2 land coverage only from 0.2 % to 0.6 %, and leave NO_2 urban coverage unchanged at 6.1 % (Section 7.3), confirming that closing NO_2 coverage gaps nationally would require substantially more sensors concentrated near traffic infrastructure rather than a broad regional deployment. The placement output is a data-driven research artefact and recommendation; it does not constitute an operational infrastructure plan.

Read individually, these three answers describe three separate analyses; read together, as traced in Table 10 (Section 7.4), they describe one causal chain per pollutant. The spatial autocorrelation structure that answers RQ1 fixes the shape of the accuracy-distance decay that answers RQ2, which in turn fixes the sensor budget required for the coverage outcome that answers RQ3. It is this chain, not any single RQ answer in isolation, that grounds the thesis's central claim: a supplementary IoT network sized and sited for one pollutant's spatial behaviour cannot be assumed to serve the other.

8.2 Comparison with Related Work

The finding that Random Forest and IDW both outperform LUR is consistent with the broader machine learning literature, which reports a consistent pattern of ML methods outperforming regression baselines on RMSE and MAE across diverse geographic contexts [12]. A direct comparison against the ESCAPE cross-validation R^2 benchmarks of 0.71 for $PM_{2.5}$ [10] and 0.83 for NO_2 [11] is not straightforward, however: the ESCAPE benchmarks are derived from spatial cross-validation of annual mean concentrations across 20–40 monitoring sites per study area, whereas the R^2 values reported in Section 5 are computed per station across daily time series under buffered SLOO, a substantially stricter and differently structured evaluation that frequently yields negative R^2 even when RMSE is low in absolute terms (Section 5). The RMSE-based comparison against the LUR benchmark, calibrated on the same station network, therefore provides the more directly comparable evaluation within this study, and both Random Forest and IDW exceed it for both pollutants, though the smaller number of stations eligible for SLOO validation relative to ESCAPE-scale studies (11 concurrently-passing stations here, compared with 20–40 monitoring sites per ESCAPE study area) constrains how far this comparison can be extended to draw firm conclusions about performance relative to the wider European literature.

The result that IDW outperforms Random Forest for $PM_{2.5}$ diverges from the general expectation in the ML literature that feature-based learning outperforms simple deterministic interpolation [13]. This divergence is attributable to the sparse, geographically dispersed configuration of the Swedish reference network: where the spatial autocorrelation range of a pollutant is long relative to inter-station spacing, as established for $PM_{2.5}$ in Section 3.2, a distance-weighted average of nearby stations captures the spatial signal effectively without the additional predictive machinery of land-use, topographic, and meteorological covariates. The pronounced urban/rural NO_2 gradient identified in Section 6.3 is consistent with high-resolution Swedish dispersion modelling confirming that substantial intra-urban exposure heterogeneity persists in contemporary Swedish cities [2].

The greedy sequential placement approach adopted in Section 3.5 in preference to Particle Swarm Optimization [16, 17] is supported by the placement results: the constant marginal coverage gain observed across all 20 steps (Section 7.2) indicates that the underlying coverage landscape is not yet multimodal at this deployment scale, the condition under which a population-based metaheuristic would be expected to offer an advantage over greedy search.

8.3 Implications for Municipal IoT Planning

The asymmetry between the $PM_{2.5}$ and NO_2 thresholds carries a direct implication for supplementary IoT network design. A sensor network sized to close $PM_{2.5}$ coverage gaps, as demonstrated in Section 7, is not a sensor network capable of resolving NO_2 exposure at comparable cost: the roughly tenfold difference between the two reliable prediction distances means that a deployment planned exclusively around the $PM_{2.5}$ criterion would leave NO_2 exposure, concentrated near traffic infrastructure, effectively unmonitored between reference stations. This is not merely a theoretical consequence of the threshold gap: the same 20 $PM_{2.5}$-sited sensors evaluated against the NO_2 threshold in Section 7.3 left national

NO_2 urban coverage completely unchanged, at 6.1 %, confirming that a $PM_{2.5}$-oriented deployment provides no incidental NO_2 benefit in practice. Municipalities and regional environmental agencies seeking to address both pollutants with a single IoT deployment should treat the two thresholds as separate planning constraints rather than a shared coverage objective.

The four administrative regions served by the recommended placements, Jönköping, Örebro, Dalarna, and Norrbotten (Section 7.3), represent concrete near-term priorities for regional environmental agencies currently operating without any concurrent-passing reference station. As stated throughout Section 7, the placement output is a data-driven recommendation and starting point for site-level planning; it does not determine specific sensor hardware, mounting locations, or regulatory compliance status, all of which require further site-level assessment.

8.4 Limitations

Eleven concurrently-passing stations are eligible to be withheld as validation targets under the buffered SLOO protocol (Section 4.1); this is substantially smaller than the 20–40 monitoring sites comprising each ESCAPE study area [10, 11], constraining the statistical power of the validation and the granularity of the resulting decay curves. The 5 km SLOO exclusion buffer is justified by the clustering structure of the Swedish network rather than by the empirical variogram range (Section 3.3); error estimates may consequently be conservative in network-dense areas and optimistic in network-sparse areas where the buffer constraint is rarely triggered. Euclidean distance is used throughout the accuracy-distance analysis in place of road-network distance, which would be more physically appropriate for NO_2 given its traffic-source dominance, but was precluded by the sparsity of the underlying network (Section 3.4). The NO_2 decay analysis is further confounded by the urban/rural composition of the station sample, with only two rural stations (Bredkälen and Norr Malma) among the 11 analytical stations (Section 6.3); the reported 6 km threshold should therefore be interpreted as an indicator of rapid NO_2 degradation rather than a precisely calibrated figure. The feature set used for spatial estimation comprises 30 predictor variables (Section 3.2) drawn from a training network of 11 unique spatial locations; this ratio is high by the standards of linear modelling, and although Random Forest's bootstrap aggregation is argued in Section 3.2 to mitigate the resulting overfitting risk, the risk itself is not eliminated and should be weighed when interpreting the reported accuracy figures. The national coverage grid used for the placement analysis (Section 7.1) is a further constraint specific to NO_2: at 10 km spacing, the grid resolution is coarser than the 6 km NO_2 reliable prediction threshold itself, so individual grid cells cannot resolve NO_2 coverage geometry at the scale the threshold implies. This is a distinct limitation from the small-sensor-budget argument given in Section 7.1: even a NO_2-targeted placement evaluated on this grid would understate true coverage in a way the $PM_{2.5}$ analysis, whose 64 km threshold is far larger than the grid spacing, does not. The 50 per cent threshold criterion itself is a further source of uncertainty: the sensitivity analysis in Section 6.4 shows that post-placement $PM_{2.5}$ coverage ranges from 2.9 % to 94.7 % across a 20 to 80 per cent sweep of the criterion, against the 67.2 % figure reported at the 50 per cent criterion adopted here, so the specific coverage percentages should be treated as contingent on this modelling choice rather than as precise estimates. Finally, the accuracy-distance relationship and the resulting thresholds are calibrated to the specific spatial distribution of the Swedish SMHI network and may not transfer directly to national networks of different density or configuration.

8.5 Risks and Constraints

The placement artefact carries a risk of misapplication if its output is interpreted as a regulatory compliance assessment or an operational deployment plan rather than the data-driven planning recommendation it is intended to be; a location falling within the reliable prediction distance of an existing station indicates that spatial estimation is trustworthy at that distance, not that measured or estimated concentrations at that location meet EU or Swedish air quality limit values. The accuracy-distance thresholds are calibrated to the 2020–2024 study period and the emission patterns prevailing during it; structural shifts in traffic composition, such as continued electrification of the vehicle fleet, could alter the spatial gradient of NO_2 in particular and would require the decay analysis to be re-run against more recent data before the thresholds are relied upon for future planning decisions. Finally, the reliability of both the estimation models and the placement artefact is contingent on the continued operation of the existing SMHI reference network: closure or relocation of any of the 11 analytical stations would alter the coverage surface underlying the placement recommendations and would require the analysis to be repeated against the updated network configuration.

9 Conclusion

This thesis investigated the accuracy with which $PM_{2.5}$ and NO_2 concentrations can be estimated at unmonitored locations from the sparse Swedish SMHI reference network, characterised how that accuracy degrades with distance from the nearest station, and used the resulting relationship to derive a data-driven criterion for supplementary IoT sensor placement. Following a Design Science Research methodology, a Random Forest model was evaluated against inverse distance weighting and land-use regression benchmarks under a buffered spatial leave-one-out cross-validation protocol, and the resulting per-station errors were used to fit accuracy-distance decay curves and derive reliable prediction distance thresholds for both pollutants. A greedy sequential placement algorithm then applied these thresholds to a national coverage grid to identify priority IoT sensor locations.

The three research questions are answered as follows.

1. **RQ1:** Under buffered SLOO cross-validation, Random Forest achieves the lowest mean RMSE for NO_2 ($7.27\,\mu\mathrm{g\,m^{-3}}$), outperforming both the LUR benchmark and the IDW baseline on this aggregate measure. For $PM_{2.5}$, IDW achieves the lowest mean RMSE ($2.33\,\mu\mathrm{g\,m^{-3}}$), outperforming Random Forest, reflecting the smoother spatial autocorrelation structure of $PM_{2.5}$ relative to NO_2. LUR is outperformed by both alternative methods for both pollutants.

2. **RQ2:** $PM_{2.5}$ estimation accuracy degrades gradually with distance, remaining reliable within approximately 64 km of the nearest reference station. NO_2 estimation accuracy degrades far more steeply, with a reliable prediction distance of approximately 6 km, reflecting the sharp, traffic-sourced spatial gradients characteristic of that pollutant.

3. **RQ3:** Applying the 64 km $PM_{2.5}$ threshold, a greedy sequential placement algorithm identifies 20 candidate IoT sensor locations that raise national $PM_{2.5}$ land-area coverage from 10.9 % to 67.2 % and urban-area coverage from 38.8 % to 71.4 %, extending reliable coverage to four of the seven Swedish administrative regions currently without a concurrent-passing reference station. RQ3 is therefore answered affirmatively for $PM_{2.5}$: the accuracy-distance relationship yields an actionable, sensor-budget-appropriate placement criterion at national scale. The same placement

heuristic is not actionable for NO_2 at this sensor budget: applied against the 6 km NO_2 threshold, the identical 20 sensors raise national NO_2 land coverage only from 0.2 % to 0.6 % and leave urban coverage unchanged at 6.1 %. RQ3 is thus effectively unanswerable for NO_2 within a sensor budget of this size; closing NO_2 coverage gaps nationally would require an order of magnitude more sensors, concentrated near traffic infrastructure, confirming that a $PM_{2.5}$-sized deployment does not generalise to NO_2.

The contribution of this thesis is an empirically grounded, reproducible framework that translates spatial estimation accuracy into a concrete, prioritised sensor placement criterion, demonstrated at national scale on the Swedish SMHI reference network. The framework combines a validated spatial estimation model, a quantified accuracy-distance relationship, and a transparent greedy placement algorithm within a single Design Science Research artefact. The work does not constitute an operational deployment plan and does not determine regulatory compliance; it provides a data-driven starting point for municipal and regional planning that can be refined through site-level assessment.

The limitations discussed in Section 8.4, principally the size of the training network relative to comparable European studies and the calibration of all thresholds to the specific spatial configuration of the Swedish network, point directly to the most promising directions for future work. Validation against a denser monitoring network, or replication in a different national context, would allow the accuracy-distance relationship to be generalised beyond the sparse station configuration used here. Extending the placement artefact to jointly optimise for both pollutants, rather than applying the $PM_{2.5}$ threshold alone, would address the coverage asymmetry identified in Section 8.3 and represents a natural next step once a denser NO_2-capable network configuration becomes available for calibration; such an extension would also require a national grid finer than the current 10 km spacing, since the grid resolution itself is coarser than the 6 km NO_2 threshold (Section 8.4). Finally, periodic re-calibration of the decay curves against more recent data would allow the placement criterion to track structural shifts in emission patterns, such as vehicle fleet electrification, over time.

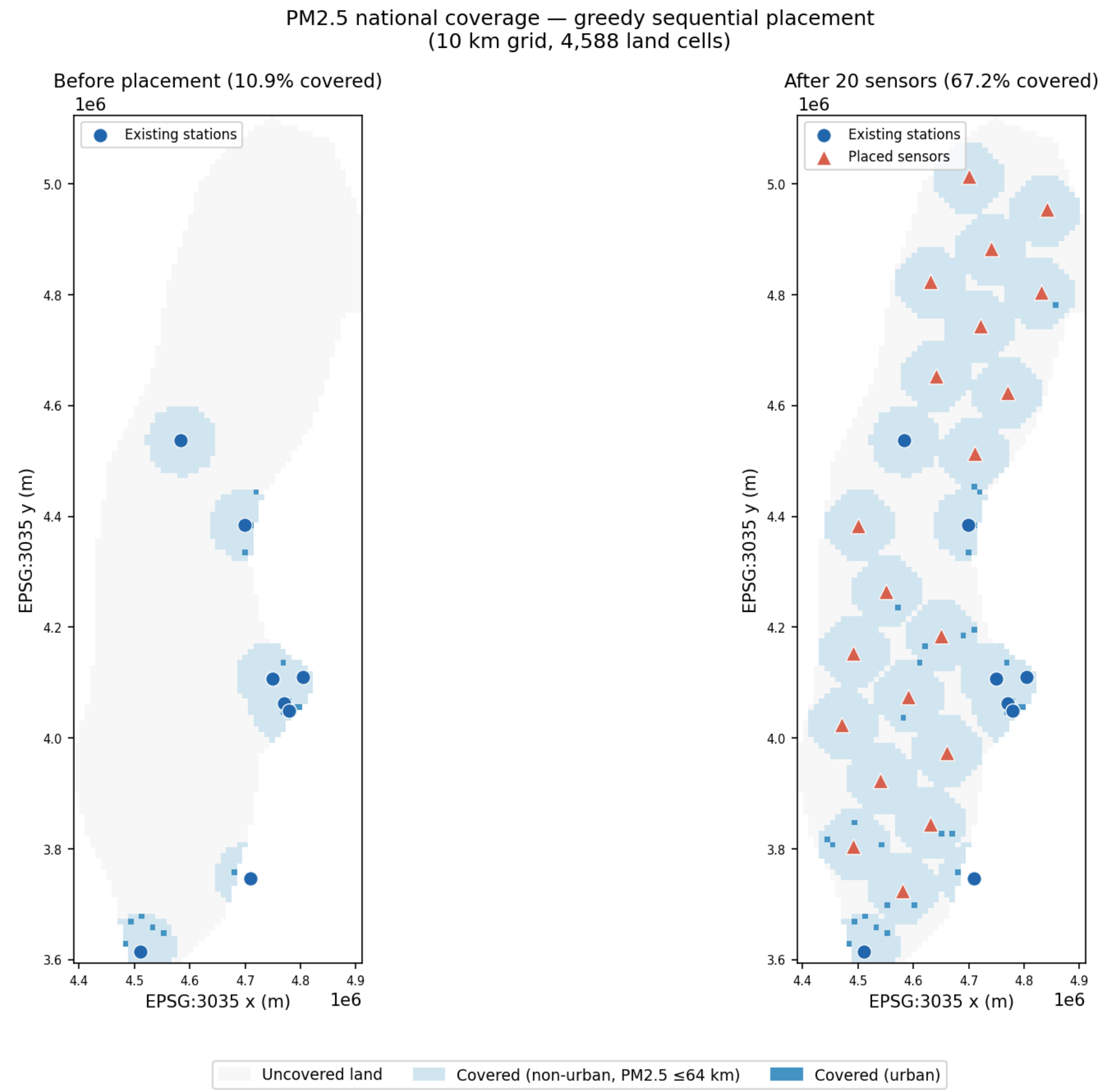

Figure 6: $PM_{2.5}$ national coverage before (left) and after (right) 20 greedy placements. Light grey cells are beyond the 64 km reliable estimation threshold; light blue cells are covered non-urban land; dark blue cells are covered urban land (CORINE 111 and 112). Red circles indicate existing stations; green triangles indicate recommended sensor locations (right panel only). Grid resolution: 10 km.

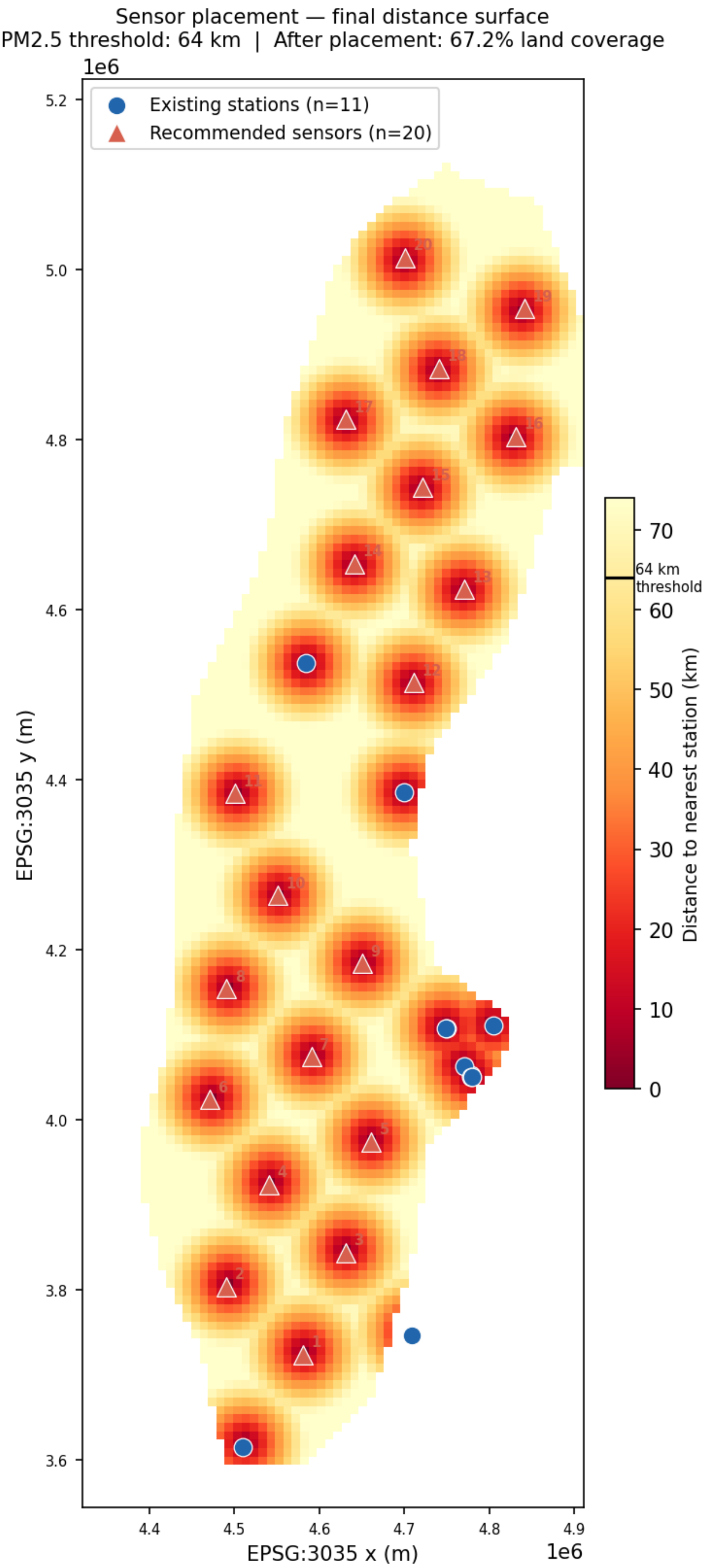

Figure 7: Distance to nearest monitoring station (km) across the Swedish national grid after greedy placement, combining the 11 existing stations and the 20 recommended sensors. The colourbar threshold at 64 marks the $PM_{2.5}$ reliable estimation boundary. Existing stations are shown as blue circles; recommended sensors as red triangles, labelled by

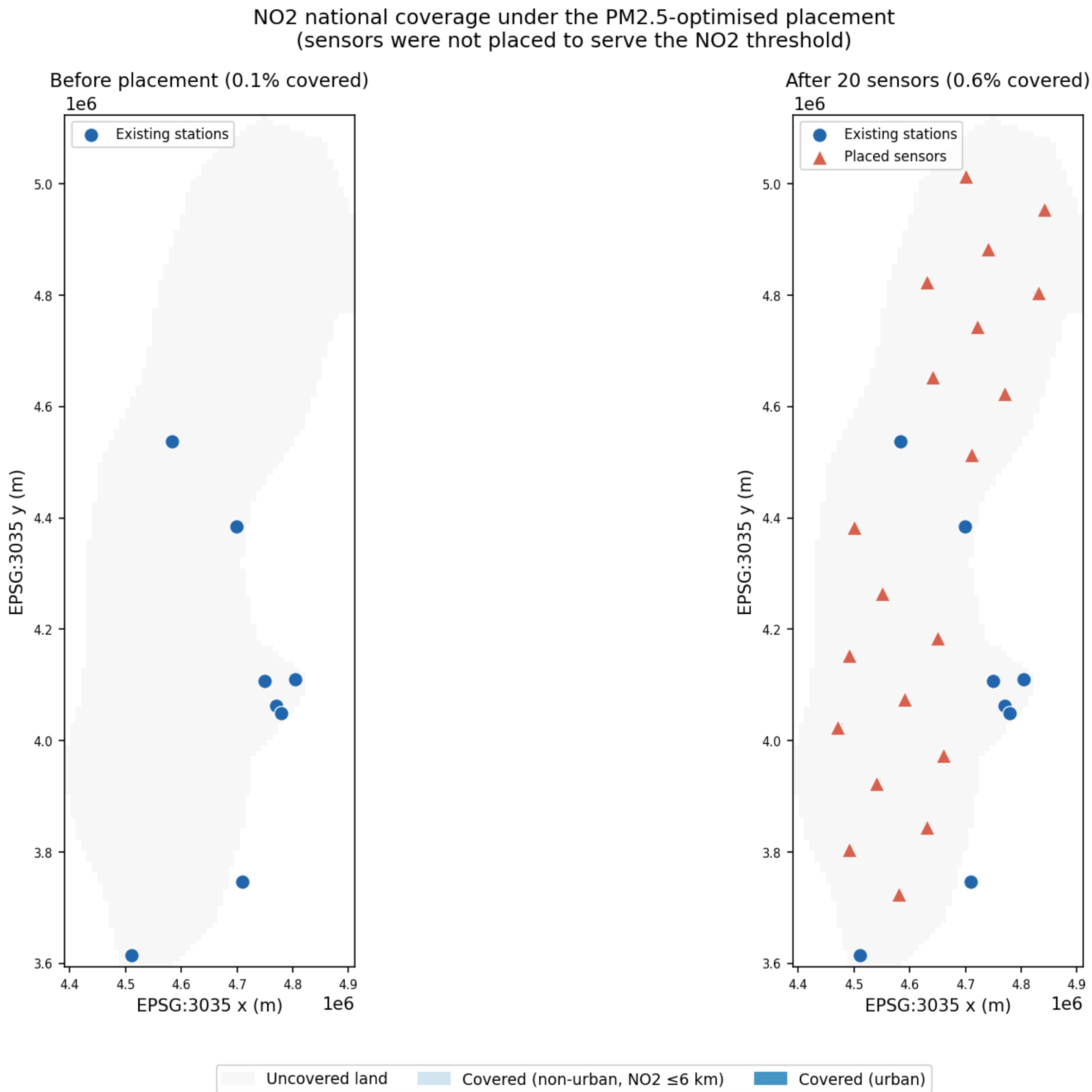

Figure 8: NO_2 national coverage before (left) and after (right) the same 20 $PM_{2.5}$-optimised placements, evaluated against the 6 km NO_2 reliable estimation threshold. Colour scheme as in Figure 6. Because the threshold is smaller than the 10 km grid spacing, covered cells appear only immediately around each station or sensor location.

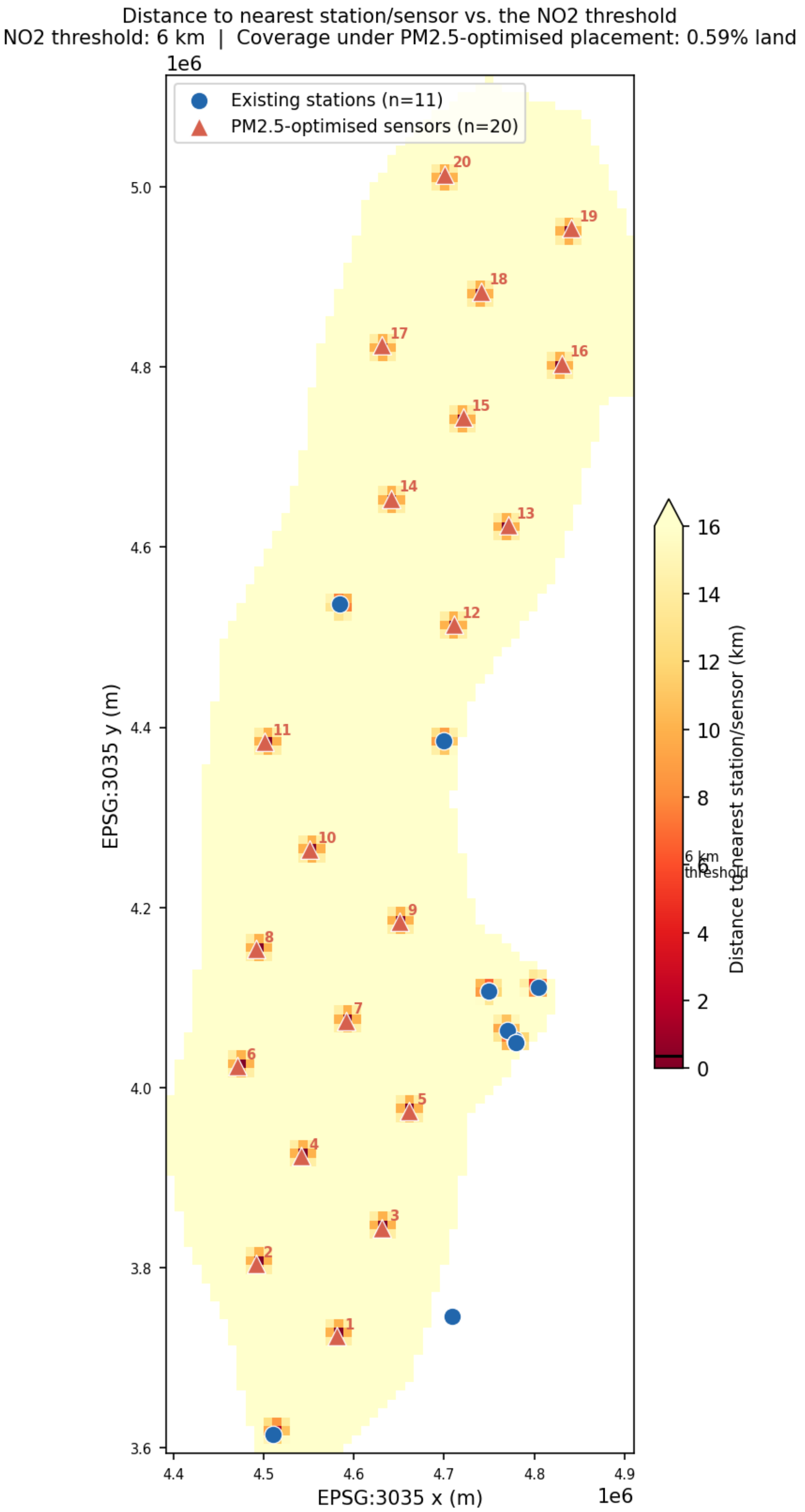

Figure 9: Distance to nearest monitoring station or sensor (km) across the Swedish national grid after the same 20 $PM_{2.5}$-optimised placements, evaluated against the 6 km NO_2 reliable estimation threshold. Existing stations are shown as blue circles; $PM_{2.5}$-optimised sensors as red triangles, labelled by placement rank. Nearly the entire grid remains beyond the NO_2 threshold.

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A Station Inventory and Completeness

Table 11 lists all 34 stations identified with $PM_{2.5}$ or NO_2 records during the 2020–2024 study period (Section 4.1), together with their per-pollutant completeness against the full 1,827-day period. *Both (analytical sample)* denotes the 11 stations meeting the 90 per cent threshold concurrently for both pollutants, used as validation targets throughout Sections 5–6 and shown individually in Tables 5 and 6. *NO_2 only* and *$PM_{2.5}$ only* denote stations meeting the threshold for one pollutant but not the other; *Neither* denotes stations meeting it for neither pollutant. Stations in all four categories remain in the feature matrix and may contribute training rows to folds in which they are not the withheld station or excluded by the SLOO buffer (Section 3.3); only the 11 concurrently-passing stations are ever withheld as validation targets, for the reason given in Section 4.1.

Table 11: Station inventory: type, coordinates, and per-pollutant completeness against the full 1,827-day study period (2020–2024).

Station	**Type**	**Lat.**	**Lon.**	**NO_2 %**	**$PM_{2.5}$ %**	**Status**
Bredkälen	Rural Background	63.8455	15.3197	96.3	97.1	Both (analytical sample)
Kalmar Södra Vägen	Urban Traffic	56.6642	16.3336	98.0	97.4	Both (analytical sample)
Malmö Rådhuset	Urban Background	55.6064	13.0020	99.6	98.1	Both (analytical sample)
Norr Malma	Rural-Regional Bg.	59.8324	18.6313	99.5	99.5	Both (analytical sample)
Sollentuna E4 Häggvik	Urban Traffic	59.4422	17.9233	99.5	99.5	Both (analytical sample)
Stockholm Hornsgatan	Urban Traffic	59.3173	18.0490	98.9	100.0	Both (analytical sample)
Stockholm St Eriksgatan	Urban Traffic	59.3406	18.0370	95.2	95.4	Both (analytical sample)
Stockholm Torkel Kn.	Urban Background	59.3160	18.0578	99.9	99.8	Both (analytical sample)
Sundsvall Köpmangatan	Urban Traffic	62.3886	17.3089	100.0	99.7	Both (analytical sample)
Uppsala Dragarbrunnsg.	Urban Background	59.8605	17.6376	98.4	98.5	Both (analytical sample)
Uppsala Kungsgatan	Urban Traffic	59.8572	17.6461	97.8	99.3	Both (analytical sample)
Göteborg Haga	Urban Traffic	57.6979	11.9605	90.5	69.5	NO_2 only
Lund Trollebergsvägen	Urban Traffic	55.7033	13.1803	97.6	1.8	NO_2 only
Malmö Dalaplan	Urban Traffic	55.5847	13.0062	100.0	58.5	NO_2 only
Råö	Rural Background	57.3937	11.9142	99.9	22.8	NO_2 only
Stockholm Lilla Essingen	Urban Traffic	59.3255	18.0044	93.5	16.1	NO_2 only
Stockholm Sveavägen	Urban Traffic	59.3408	18.0583	98.5	14.2	NO_2 only
Östersund Rådhusgatan	Urban Traffic	63.1739	14.6411	94.8	79.1	NO_2 only

Table 11 continued

Station	Type	Lat.	Lon.	NO_2 %	$PM_{2.5}$ %	Status
Umeå Västra Esplanaden	Urban Traffic	63.8289	20.2585	85.3	98.9	$PM_{2.5}$ only
Botkyrka Kumla g.v. syd	Suburban Traffic	59.2366	17.8362	36.3	36.6	Neither
Gävle Staketgatan	Urban Traffic	60.6762	17.1376	59.9	59.4	Neither
Härnösand Storgatan	Urban Traffic	62.6299	17.9358	7.4	35.4	Neither
Karlstad Jungmansgatan	Urban Traffic	59.3767	13.4903	58.1	33.3	Neither
Linköping Hamngatan	Urban Traffic	58.4124	15.6301	59.7	59.9	Neither
Norrköping Kungsgatan	Urban Traffic	58.5915	16.1779	79.4	79.4	Neither
Norrköping Trädgårdsg.	Urban Background	58.5920	16.1893	71.4	74.9	Neither
Solna Enköpingsvägen	Urban Traffic	59.3769	18.0060	10.8	11.0	Neither
Solna Råsundavägen	Urban Traffic	59.3651	17.9969	84.1	84.2	Neither
Sundbyberg Tulegatan	Urban Traffic	59.3643	17.9763	77.2	77.3	Neither
Timrå	Urban Background	62.4868	17.3244	9.9	9.9	Neither
Varberg Västra Vallg.	Urban Traffic	57.1055	12.2485	19.2	20.0	Neither
Västerås Melkertorget	Urban Traffic	59.6099	16.5505	41.9	40.0	Neither
Västerås Stora Gatan	Urban Traffic	59.6077	16.5359	57.1	51.9	Neither
Växjö Liedbergsgatan	Urban Traffic	56.8799	14.7993	33.3	56.8	Neither

Of the 34 stations, 11 pass the 90 per cent threshold concurrently for both pollutants, 7 pass for NO_2 only, 1 passes for $PM_{2.5}$ only, and 15 pass for neither pollutant. The 8 single-pollutant passers are not used as validation targets for either pollutant, for the comparability reason given in Section 4.1: using a different validation station set per pollutant would confound genuine differences in accuracy-distance behaviour between $PM_{2.5}$ and NO_2 with differences in the geographic sampling of the two validation networks.

B Model Hyperparameters

Tables 12–14 document the configuration of the three estimators evaluated in Section 5, as implemented for the buffered SLOO procedure (Section 3.3). All three are fitted independently within each fold, using only the training stations retained after the 5 km buffer exclusion.

Preprocessing. Random Forest and LUR both consume the 30-feature predictor matrix described in Section 3.2. Any feature values still missing after the precipitation imputation described in Section 4.7 are imputed a second time using the median of each feature column, computed on the training fold only and applied unchanged to the held-out station, to avoid leakage across the buffer boundary. IDW does not use the feature matrix: it operates solely on inter-station distances and the raw (non-imputed) target values available on each day.

C Placement Algorithm Parameters

Table 15 documents the parameters used in the greedy sequential placement algorithm described in Section 3.5 and applied in Section 7.2.

Table 12: Random Forest configuration (scikit-learn `RandomForestRegressor`).

Parameter	**Value**	**Note**
Number of trees (`n_estimators`)	200	Fixed across all folds and both pollutants
Minimum samples per leaf (`min_samples_leaf`)	5	Only explicit regularisation applied; limits overfitting to individual daily observations
Maximum tree depth	None (default)	Trees grown until leaves reach `min_samples_leaf`
Features considered per split	All 30 (default for regression)	No feature subsampling
Bootstrap sampling	Enabled (default)	Standard bagging
Random seed (`random_state`)	42	Fixed for reproducibility across folds

Table 13: Land-use regression (LUR) configuration (scikit-learn `LinearRegression`).

Parameter	**Value**	**Note**
Estimator	Ordinary least squares	No regularisation (ridge/lasso) applied
Intercept	Fitted (default)	
Feature scaling	None	Raw imputed feature values used directly
Predictors	Same 30 features as RF	Section 3.2

Table 14: Inverse distance weighting (IDW) configuration.

Parameter	**Value**	**Note**
Distance power	2	Weight $\propto 1/d^2$; standard IDW exponent
Distance metric	Haversine (km)	From the inter-station distance matrix (Section 3.3)
Neighbour set	All retained training stations	No fixed k; on each day, only training stations with a non-null observation that day contribute
Predictors used	None	Distance and target concentration only

Table 15: Greedy sequential sensor placement: algorithm parameters and justifications.

Parameter	**Value**	**Justification**
Grid resolution	10 km (EPSG:3035)	Computationally tractable; coarser than site-level precision but adequate for regional gap identification
$PM_{2.5}$ coverage threshold	64 km	Derived from IDW exponential decay model (Section 6.4)
NO_2 coverage threshold	6 km	Derived from IDW log decay model (Section 6.4); reported as secondary metric only
Optimisation objective	$PM_{2.5}$ land area coverage	NO_2 threshold (6 km) is not actionable at national scale with a small sensor budget
Maximum steps	20	Covers the non-overlapping placement horizon under current network density
Land mask	EU-DEM + Sweden polygon	DEM excludes ocean; polygon restricts to Swedish mainland, excluding Norwegian and Finnish territory
Urban classification	CORINE 111 + 112	Continuous and discontinuous urban fabric; secondary evaluation metric
Distance metric	Euclidean (EPSG:3035)	Valid approximation for distances up to 200 km at Swedish latitudes